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Galvin IM, Levy R, Boyd JG, Day AG, Wallace MC.
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Cochrane Database of Systematic Reviews 2015, Issue 1. Art. No.: CD006638.
DOI: 10.1002/14651858.CD006638.pub3.

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[Intervention Review]

Cooling for cerebral protection during brain surgery

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Editorial group: Cochrane Anaesthesia Group.
Publication status and date: New search for studies and content updated (no change to conclusions), published in Issue 1, 2015.

Citation: Galvin IM, Levy R, Boyd JG, Day AG, Wallace MC. Cooling for cerebral protection during brain surgery. *Cochrane Database of Systematic Reviews* 2015, Issue 1. Art. No.: CD006638. DOI: 10.1002/14651858.CD006638.pub3.

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A B S T R A C T

Background

Patients undergoing neurosurgery are at risk of cerebral ischaemia with resultant cerebral hypoxia and neuronal cell death. This can increase both the risk of mortality and long term neurological disability. Induced hypothermia has been shown to reduce the risk of cerebral ischaemic damage in both animal studies and in humans who have been resuscitated following cardiac arrest. This had lead to an increasing interest in its neuroprotective potential in neurosurgical patients. This review was originally published in 2011 and did not find any evidence of either effectiveness or harm in these patients. This updated review was designed to capture current evidence to readdress these issues.

Objectives

To evaluate the effectiveness and safety profile of induced hypothermia versus normothermia for neuroprotection in patients undergoing brain surgery. Effectiveness was to be measured in terms of short and long term mortality and functional neurological outcomes. Safety was to be assessed in terms of the rate of the adverse events infection, myocardial infarction, ischaemic stroke, congestive cardiac failure and any other adverse events reported by the authors of the included studies.

Search methods

For the original review, the authors searched the databases Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE (OvidSP), EMBASE (OvidSP) and LILACS to November 2010. For the updated review all these databases were re-searched from November 2010 to May 2014.

For both the original and updated versions, grey literature was sought by searching reference lists of identified studies and relevant review articles, and conference proceedings. No language restrictions were applied.

Selection criteria

As in the original review, we included randomized controlled trials (RCTs) of induced hypothermia versus normothermia for neuroprotection in patients of any age and gender undergoing brain surgery, which addressed mortality, neurological morbidity or adverse event outcomes.

Data collection and analysis

Three review authors independently extracted data and two independently assessed the risk of bias of the included studies. Any discrepancies were resolved by discussion between authors.

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Main results

In this updated review, one new ongoing study was found but no new eligible completed studies were identified. This update was therefore conducted using the same four studies included in the original review. These studies included a total of 1219 participants, mean age 40 to 54 years. All included studies were reported as RCTs. Two were multicentred, together including a total of 1114 patients who underwent cerebral aneurysm clipping, and were judged to have an overall low risk of bias. The other two studies were single centred. One included 80 patients who had a craniotomy following severe traumatic brain injury and was judged to have an unclear or low risk of bias. The other study included 25 patients who underwent hemicranicectomy to relieve oedema following cerebral infarction and was judged to have an unclear or high risk of bias. All studies assessed hypothermia versus normothermia. Overall 608 participants received hypothermia with target temperatures ranging from 32.5 °C to 35 °C, and 611 were assigned to normothermia with the actual temperatures recorded in this group ranging form 36.5 °C to 38 °C. For those who were cooled, 556 had cooling commenced immediately after induction of anaesthesia that was continued until the surgical objective of aneurysm clipping was achieved, and 52 had cooling commenced immediately after surgery and continued for 48 to 96 hours.

Pooled estimates of effect were calculated for the outcomes mortality during treatment or follow-up (ranging from in-hospital to one year); neurological outcome measured in terms of the Glasgow Outcome Score (GOS) of 3 or less; and adverse events of infections, myocardial infarction, ischaemic stroke and congestive cardiac failure. With regards to mortality, the risk of dying if allocated to hypothermia compared to normothermia was not statistically significantly different (risk ratio (RR) 0.87, 95% confidence interval (CI) 0.59 to 1.27, P = 0.47). There was no indication that the time at which cooling was started affected the risk of dying (RR with intraoperative cooling 0.95, 95% CI 0.60 to 1.51, P = 0.83; RR for cooling postoperatively 0.67, 95% CI 0.34 to 1.35, P = 0.26). For the neurological outcome, the risk of having a poor outcome with a GOS of 3 or less was not statistically different in those who received hypothermia versus normothermia (RR 0.80, 95% CI 0.61 to 1.04, P = 0.09). Again there was no indication that the time at which cooling was started affected this result. Regarding adverse events, there was no statistically significant difference in the incidence in those allocated to hypothermia versus normothermia for risk of surgical infection (RR 1.20, 95% CI 0.73 to 1.97, P = 0.48), myocardial infarction (RR 1.86, 95% CI 0.69 to 4.98, P = 0.22), ischaemic stroke (RR 0.93, 95% CI 0.82 to 1.05, P = 0.24) or congestive heart failure (RR 0.85, 95% CI 0.60 to 1.21, P = 0.38). In contrast to other outcomes, where time of application of cooling did not change the statistical significance of the effect estimates, there was a weak statistically significant increased risk of infection in those who were cooled postoperatively versus those who were not cooled (RR 1.77, 95% CI 1.05 to 2.98, P = 0.03). Overall, as in the original review, no evidence was found that the use of induced hypothermia was either beneficial or harmful in patients undergoing neurosurgery.

Authors' conclusions

We found no evidence that the use of induced hypothermia was associated with a significant reduction in mortality or severe neurological disability, or an increase in harm in patients undergoing neurosurgery.

PLAIN LANGUAGE SUMMARY

No evidence that cooling patients undergoing brain surgery helps to either reduce death and severe disability or increase harm

Brain surgery is often required in patients who have abnormal brain blood vessels, had a stroke, sustained a traumatic injury or have suffered spontaneous bleeding into the brain. While surgery can be lifesaving it can also cause further damage due to an associated interruption and reduction of blood flow to the brain cells. Without an adequate supply of blood the brain cells are deprived of oxygen and they may die as a result. If brain cells are lost in large numbers or in important areas of the brain, the patient may die or be severely disabled. This has led to a search for ways to protect the brain cells during brain operations. Studies in animals have shown that lowering the body temperature can help to protect the brain cells in circumstances when the blood supply to the cells is compromised. Similarly, studies in humans who have been resuscitated after their heart stopped beating have shown that lowering their body temperature helps to reduce brain damage. The purpose of this updated version of a previous review of the same title was to determine whether cooling patients who were having brain surgery reduced death and serious disability, or was associated with increased risk of harm. A detailed search of the available literature up until May 2014 identified four eligible studies that included a total of 1219 participants. Their results were combined to answer these questions. No evidence was found that cooling patients who are having brain operations reduced the risk of death or severe disability, or produced significant harm.

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SUMMARY OF FINDINGS

Summary of findings for the main comparison. Cooling for cerebral protection during brain surgery

Cooling for cerebral protection during brain surgery

Patient or population: cerebral protection during brain surgery
Settings: inpatient
Intervention: cooling

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of participants (studies)	Quality of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Control	Cooling				
Mortality rate Risk ratio Follow-up: 0 to 1 years	**Study population**		**RR 0.87** (0.59 to 1.27)	1216 (4 studies)	⊕⊕⊕⊖ **moderate** [1]	
	82 per 1000	**71 per 1000** (48 to 104)				
	Moderate					
	109 per 1000	**95 per 1000** (64 to 138)				
Glasgow Outcome Score (GOS) Clinical assessment Follow-up: 0 to 1 years	**Study population**		**RR 0.80** (0.61 to 1.04)	1216 (4 studies)	⊕⊕⊕⊖ **moderate** [1]	
	170 per 1000	**136 per 1000** (104 to 177)				
	Moderate					
	167 per 1000	**134 per 1000** (102 to 174)				
Ischaemic stroke - cooled during Clinical examination supported by imaging evidence Follow-up: 0 to 6 months	**Study population**		**RR 0.93** (0.82 to 1.05)	1114 (2 studies)	⊕⊕⊕⊕ **high**	
	487 per 1000	**453 per 1000** (400 to 512)				
	Moderate					
	365 per 1000	**339 per 1000**				

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		(299 to 383)			
Infections Unstated means	**Study population**		**RR 1.2** (0.73 to 1.97)	1194 (3 studies)	⊕⊕⊝⊝ **low** [2,3]
	311 per 1000	**373 per 1000** (227 to 613)			
	Moderate				
	325 per 1000	**390 per 1000** (237 to 640)			
Congestive heart failure - cooled during	**Study population**		**RR 0.85** (0.6 to 1.21)	1114 (2 studies)	⊕⊕⊕⊕ **high**
	108 per 1000	**91 per 1000** (65 to 130)			
	Moderate				
	60 per 1000	**51 per 1000** (36 to 73)			
Myocardial infarction	**Study population**		**RR 1.86** (0.69 to 4.98)	1139 (3 studies)	⊕⊕⊕⊝ **moder-ate** [1]
	11 per 1000	**20 per 1000** (7 to 52)			
	Moderate				
	10 per 1000	**19 per 1000** (7 to 50)			

*The basis for the **assumed risk** (e.g. the median control group risk across studies) is provided in footnotes. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).
CI: Confidence interval; **RR:** Risk ratio

GRADE Working Group grades of evidence
High quality: Further research is very unlikely to change our confidence in the estimate of effect.
Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.
Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
Very low quality: We are very uncertain about the estimate.

[1] One of the studies that contributed to this result had a high or unclear risk of bias.

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[2] There was considerable statistical heterogeneity across studies. Additionally the lack of information regarding how infection was defined and diagnosed left the potential for substantial differences in assessment and consequently a high likelihood of important clinical heterogeneity.
[3] Numbers were small and CIs were not reported.

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BACKGROUND

Description of the condition

When cerebral blood flow is disrupted due to a traumatic, haemorrhagic or ischaemic insult, anoxic cell membrane depolarization occurs. Neuronal depolarization opens calcium and sodium channels to allow these cations into the cells (Rothman 1995). The resultant osmotic gradient pulls chloride and water into the cells causing cellular swelling and breakdown. The hypoxic cerebral cells also release glutamate, leading to an increase in intracellular calcium levels and further cell damage (Choi 1990). As part of a generalized and localized inflammatory response oxygen free radicals are produced, which cause mitochondrial dysfunction and further neuronal cell necrosis and apoptosis. When the cerebral blood flow is restored further damage can occur due to a secondary superoxide surge, increased production of nitric oxide and disruption of the blood brain barrier (Wei 1999). This phase is termed reperfusion injury and can lead to even more severe damage than the initial hypoxic insult.

Patients who have sustained traumatic brain injury, intracerebral haemorrhage or ischaemia often require surgical intervention, as do those who require repair of vascular malformations. Surgery carries the potential for further cerebral ischaemia due to interruption of cerebral perfusion, vasospasm, impaired autoregulation and hyperaemic reperfusion injury, further compromising clinical outcomes (Leslie 2003). This has lead to a search for neuroprotective strategies to limit the extent of surgical induced damaged (El Beheiry 2012).

Description of the intervention

The use of hypothermia in neurosurgery was first described in the 1950s, with profound hypothermia being postulated as having neuroprotective effects (Hendrick 1959; Lougheed 1955). However it was largely abandoned due to excessive complications and poor outcomes (Kwak 1972; Michenfelder 1968; Steen 1980). There was a resurgence of interest in this area in the 1990s with studies showing that mild hypothermia improved outcomes in animal models of ischaemic and traumatic brain injury (Clifton 1991; Goto 1993; Karibe 1994). Clinical studies in post-cardiac arrest populations showed improved outcomes, triggering further interest in its potential neuroprotective effects in ischaemic stroke, traumatic brain injury, cerebral aneurysm surgery etc (Els 2006; Holzer 2002; Qiu 2007; Todd 2005). Common to all of these conditions is the disruption of cerebral blood flow with resultant ischaemic cell death.

How the intervention might work

Hypothermia protects the brain during hypoxic injury by a reduction in cerebral metabolism by approximately 5% for every degree reduction in temperature, attenuating or delaying anoxic depolarization, and reducing glutamate release (Laptook 1995). It also reduces the production of excitatory amino acids and suppresses nitric oxide and superoxide production (Drury 2014). In vitro studies have shown that mild hypothermia reduces neuronal apoptosis by reducing the production and activation of hypoxia associated proteins that mediate cell death (Xu 2002). Animal studies have shown that hypothermia also attenuates the mechanisms involved in reperfusion injury (Thoresen 1997).

Why it is important to do this review

Recently there has been increasing focus on the role of hypothermia in those people undergoing neurosurgical procedures including aneurysm repair, haematoma evacuation, and decompressive craniectomy (Els 2006; Hindman 1999; Qiu 2007). The original review (Milani 2011) of the value of hypothermia in neurosurgical populations was carried out at the beginning of this emerging focus of research. It aimed to provide an insight into the overall benefits among and across neurosurgical populations, provide evidence based guidance on the optimal temperature targets and the timing and duration of treatment as well as its safety profile. That review incorporated existing evidence up until November 2010. This updated review was conducted to determine what the current state of evidence is in this regard.

OBJECTIVES

To evaluate the effectiveness and safety profile of induced hypothermia versus normothermia for neuroprotection in patients undergoing brain surgery. Effectiveness was to be measured in terms of short and long term mortality and functional neurological outcomes. Safety was to be assessed in terms of the rate of the adverse events infection, myocardial infarction, ischaemic stroke, congestive cardiac failure and any other adverse events reported by the authors of the included studies.

METHODS

Criteria for considering studies for this review

Types of studies

Similar to the original review on this topic (Milani 2011), we intended to include both randomized and quasi-randomized controlled trials.

Quasi-randomized trials were defined as trials using a method of allocation of participants to different forms of care that was not truly random (for example allocation by date of birth, day of the week, medical record number, month of the year, or the order in which participants were included in the study).

Types of participants

Any patient undergoing brain surgery (vascular, tumour, brain injury, intracranial arteriovenous malformations, etc), without age or gender restrictions, who were participants in a trial of cooling at any time in the perioperative period, that is either preoperatively, intraoperatively or postoperatively.

Types of interventions

Neurosurgical procedures where a deliberate strategy to cool the brain in the perioperative period was compared to no specific attempt to manipulate the brain temperature. While there is no generally accepted classification for levels of hypothermia, we defined the target levels of hypothermia as follows (where temperatures refer to brain or core values).

- Mild: down to 32 °C.
- Moderate: from 32 °C to 28 °C.
- Deep: under 28 °C.

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Types of outcome measures

Studies were eligible for inclusion if they reported any of the following outcome measures.

Primary outcomes

1. Mortality rate during treatment or the follow-up period.
2. Glasgow Outcome Score (GOS) of 3 or below. This cut off point was chosen as it marks the transition point between recovery to independent or dependent living and worse outcomes requiring high level care. Those with a GOS of 4 or more recover normal or near normal independence with employment possible with either no or some special provisions, while a GOS of 3 or less signifies severe disability, partial or total loss of independence or even death (Jennett 1975) (Appendix 1).

Secondary outcomes

The following adverse events were reported on.

1. Any surgical infective complications including cerebral infection, septicaemia, pulmonary infection etc.
2. Myocardial infarction, defined as a raised cardiac troponin level accompanied by at least one of the following: ischaemic symptoms, development of pathological Q waves on the electrocardiogram (ECG), ECG changes indicating ischaemia (ST segment elevation or depression) (JESC/ACC Committee 2000).
3. Ischaemic stroke, defined as a persistent focal neurological deficit of presumed ischaemic origin lasting more than 24 hours and confirmed by computed tomography (CT) or magnetic resonance imaging (MRI) (Goldstein 1994).
4. Congestive heart failure, defined as clinical signs of myocardial failure and requiring medical treatment.
5. Intracranial haemorrhage, defined as bleeding into the brain caused by the rupture of cerebral blood vessels (including extradural or subdural haemorrhage) confirmed by CT or MRI.
6. Any other adverse events which were reported by the authors of the primary studies.

Search methods for identification of studies

Electronic searches

For the original review (Milani 2011), the authors searched the Cochrane Central Register of Controlled Trials (CENTRAL) in *The Cochrane Library* (2010, Issue 10); MEDLINE (OvidSP) (1950 to November 2010); EMBASE (OvidSP) (1980 to November 2010); LILACS (1982 to November 2010) and CINAHL (1982 to November 2010).

For the updated review we searched all of the above databases from November 2010 to May 2014. The same search terms used in the original review were used in the updated version (CENTRAL, Appendix 2; MEDLINE, Appendix 3; EMBASE, Appendix 4; LILACS, Appendix 5; CINAHL, Appendix 6).

Searching other resources

For both the original and updated versions, grey literature was sought by searching reference lists of identified studies and relevant review articles, and conference proceedings. No language restrictions were applied.

Data collection and analysis

Selection of studies

For this updated version, three authors (IMG, RL, JGB) independently reviewed the articles retrieved by the updated search from November 2010 to May 2014 to identify studies that may have been eligible for inclusion. One review author (IMG) was responsible for contacting primary study authors should additional details have been required for determination of study eligibility and the quality. For the original version, two of the original authors scanned records retrieved by the initial search to identify potentially eligible studies and three of the original authors independently retrieved and reviewed all full-text articles (Milani 2011).

Data extraction and management

For this updated version a new data extraction form was designed by one author (IMG) (Appendix 7). The form was designed to extract data from any new studies which may be identified as well as to re-extract data from the studies included in the original review. This approach was taken to provide the reader with more comprehensive information about study methods, allow a more detailed assessment of risk of bias and ensure consistency of data extraction for all eligible studies. Three authors (IMG, RL, JGB) independently extracted the data using this form. All items recommended in Section 7 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011) were addressed in the data extraction process. Extracted data were entered initially onto the data extraction form and then into the corresponding fields in the Review Manager software (RevMan 5.2).

Assessment of risk of bias in included studies

The risk of bias of the included studies was assessed independently by two review authors (IMG, RL) using the Cochrane Collaboration tool for assessing risk of bias, as described in Section 8 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011). Any discrepancies were resolved by discussion with a third review author (JGB). For each included primary study, bias was assessed in seven domains:

1. random sequence generation,
2. allocation concealment,
3. performance bias,
4. detection bias,
5. attrition bias,
6. reporting bias, and
7. other bias.

For each domain, we determined the risk of bias as 'low', 'unclear' or 'high' according to the methods used in the study to ensure the minimization of each form of bias. In general these ratings were applied as follows.

- The low risk rating was given where information was available that clearly demonstrated that efforts were made to ensure minimal bias in that domain and the described methods were robust enough to have a high likelihood of being effective.
- The unclear risk rating was given when the information that was available was insufficient to be confident that the method used to minimize bias was robust enough to be effective.

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- The high risk rating was given when the study did not report any method to minimize bias in that domain.

Measures of treatment effect

Categorical outcomes were presented as risk ratios (RR) with 95% confidence intervals (CI). No continuous or time to event outcome measures were addressed as part of this review.

Unit of analysis issues

Data were analysed using participants who were individually randomized to each intervention group for each outcome addressed.

Dealing with missing data

For each included trial (where such information was available) the proportion of participants who were missing from the final analysis was recorded. An intention-to-treat (ITT) analysis was used, where possible.

Assessment of heterogeneity

Clinical heterogeneity was assessed by comparing study participants, settings, inventions and ancillary treatments. Methodologic heterogeneity was assessed by comparing the risk of bias in the included studies. Where included studies were deemed to be adequately homogeneous in these respects, statistical heterogeneity was assessed using the Chi^2 test and the I^2 statistic. A P value less than 0.10 and I^2 value in excess of 50% were taken as indicative of important heterogeneity (Higgins 2003).

Assessment of reporting biases

To determine the presence or absence of reporting bias, we planned to examine funnel plots for each meta-analysis that included 10 or more studies to determine if they were symmetrical. As only four studies were eligible for inclusion in this updated review an accurate assessment of reporting bias was not possible on this occasion but is planned for future updates if adequate numbers (10 or more) of eligible studies become available.

Data synthesis

Meta-analysis was conducted by the authors of the original review. RevMan (RevMan 5.2) was used to perform the analyses for each outcome in the present review. For each outcome, all eligible studies that reported the outcome were included and participants who were treated with hypothermia were compared with those who were in the normothermia group. A fixed-effect model was used in the absence of important heterogeneity and a random-effects model if heterogeneity was deemed likely (DerSimonian 1986).

Subgroup analysis and investigation of heterogeneity

Should adequate numbers of eligible studies (at least two for each subgroup) have been identified, we planned to conduct the following subgroup analyses.

- Outcomes by different types of participants, such as ischaemic stoke; haemorrhagic stoke; aneurysmal subarachnoid haemorrhage; epilepsy etc.
- Outcomes by different age groups, i.e. children versus adults.
- Outcomes by different cooling periods, i.e. preoperatively, intraoperatively and postoperatively.

Due to the small number of eligible studies retrieved, only one subgroup analysis was performed comparing outcomes in terms of mortality, GOS, surgical infection and myocardial infarction for those cooled during surgery versus those cooled after surgery. The remaining subgroup analyses are still planned for future updates of this review, depending on availability of adequate numbers (at least two for each subgroup) of eligible studies.

Sensitivity analysis

Two sensitivity analyses were performed as follows.

- Analysis excluding those studies deemed to have a low methodologic quality as evidenced by a high risk of bias.
- Analysis assuming two opposite scenarios for missing data. In the 'best case' scenario all missing patients in the hypothermia group were assumed to have a good outcome while all those in the normothermic group were assumed to have bad recovery. In the 'worst case' scenario the reverse assumption was made.

RESULTS

Description of studies

Results of the search

A total of 7272 citations were identified (5870 by the original search (Milani 2011) and 1402 by the updated search). Of these 7272 citations, a total of 7014 were excluded on initial screening as it was obvious from the titles or abstracts that they were ineligible. A total of 258 abstracts were examined further for eligibility. Of these, one new ongoing study was identified (Nuegebauer 2013), 253 were excluded, and four were included in this review (Figure 1).

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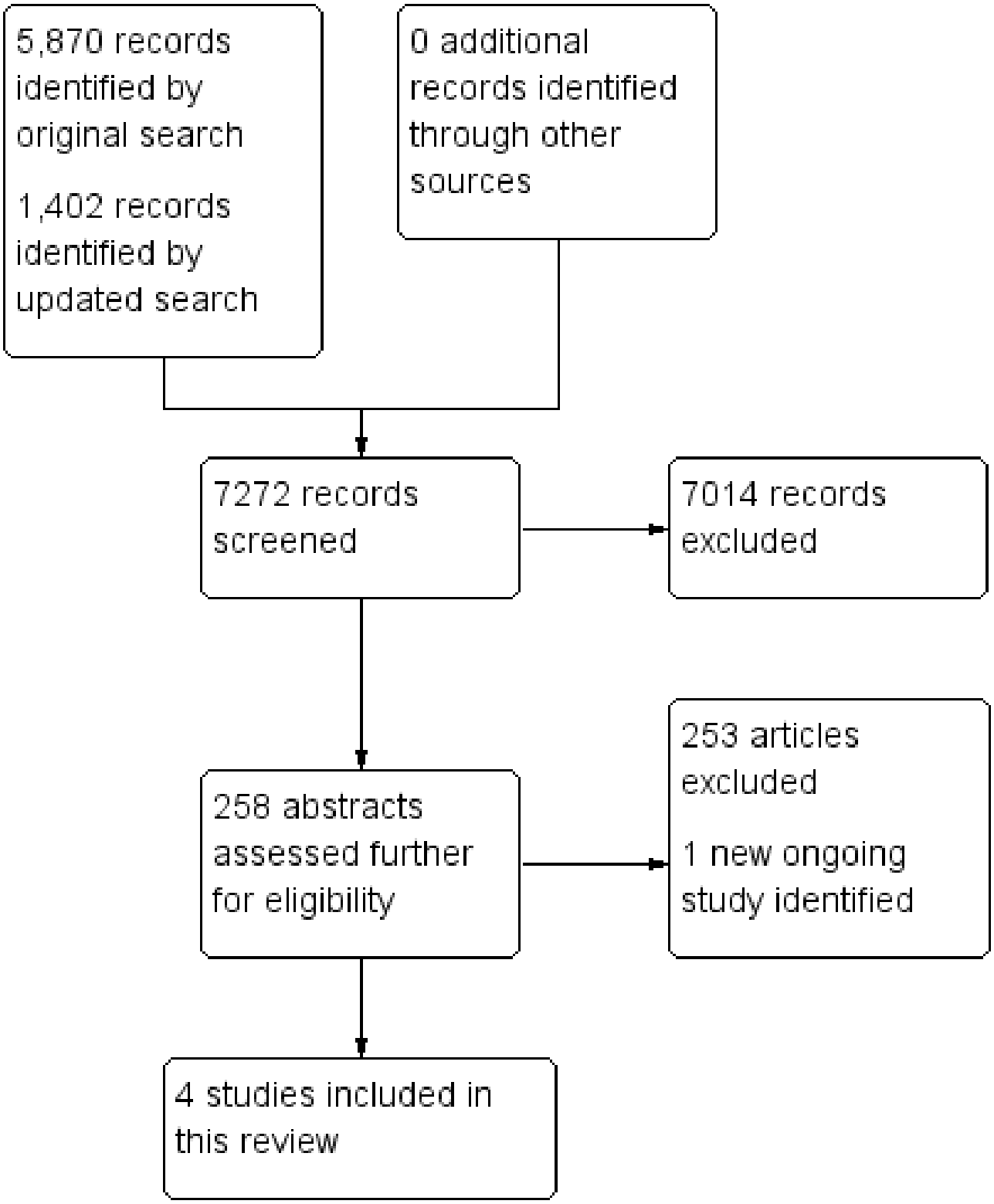
Figure 1. Study flow diagram.



Articles excluded at the title or abstract stage included animal studies, studies of interventions other then hypothermia, studies in non-surgical populations and in non-neurosurgical populations, review articles or commentaries. Articles excluded at the full-text examination stage included case series, studies evaluating non-clinical outcomes, and studies in which some participants did not undergo neurosurgery (Characteristics of excluded studies).

Included studies

Four eligible studies were included in the review.

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- Els (Els 2006) enrolled patients undergoing hemicraniectomy to treat oedema after malignant supratentorial infarction of more than two thirds of one cerebral hemisphere.
- Hindman and Todd (Hindman 1999; Todd 2005) enrolled patients undergoing cerebral aneurysm clipping.
- Qiu (Qiu 2007) enrolled patients undergoing unilateral craniotomy for operative evacuation of intracranial haematoma or massive cerebral contusion, or both, in patients with non-penetrating traumatic brain injury.

Design

All the included studies were reported to be randomized controlled trials. All studies with the exception of Els (Els 2006) reported the randomization process used.

Setting

Two studies (Hindman 1999; Todd 2005) were multicentre studies. Hindman (Hindman 1999) was performed in five academic medical centres and Todd (Todd 2005) enrolled participants from 30 centres. The other two studies (Els 2006; Qiu 2007) were conducted at single university centres.

Participants

These trials included a total of 1219 participants with the following characteristics.

Hindman (Hindman 1999) included 114 participants (9.4% of the total), 52 participants with acute subarachnoid haemorrhage (SAH) and 62 without SAH. The mean age was 51 years in the hypothermic group and 54 years in the normothermic group; 38 participants were men and 76 were women.

Todd (Todd 2005) included 1000 patients (82% of the total) who had SAH no more than 14 days before planned surgical aneurysm clipping. The median age was 52 years in the hypothermic group and 53 years in the normothermic group; 344 participants were men and 655 were women.

All of the participants in the studies by Hindman and Todd (Hindman 1999; Todd 2005) had a preoperative World Federation of Neurological Surgeons (WFNS) score (Teasdale 1988) (Appendix 8) of I, II or III. This clinical grading system, proposed by the WFNS, is based on the Glasgow Coma Score (Teasdale 1974) and the finding of a motor deficit in patients with SAH. Patients classified as grade I have the best prognosis; those with grade II have a fair prognosis; and patients with grade III tend to have poor recovery. Patients with grade IV have poor prognosis and those with grade V are classified as moribund patients.

Els (Els 2006) included 25 severely ill participants (2.1% of the total) operated on following a supratentorial ischaemic event with infarction of cerebral tissue. The mean age was 48.9 years; 15 participants were men and 10 were women.

Qiu (Qiu 2007) assessed 80 participants (6.6% of the total) who had non-penetrating severe traumatic brain injury (TBI). The mean age was 41.3 years in the hypothermic group and 40.2 years in the normothermic group; 52 participants were men and 28 were women.

Interventions

All four included studies (Els 2006; Hindman 1999; Qiu 2007;Todd 2005) assessed hypothermia versus normothermia for brain surgery.

Hindman (Hindman 1999) induced hypothermia after the start of anaesthesia and the positioning of the patient, with a target temperature of 33.5 °C. The initial operating room temperature was 20 °C to 22.0 °C. Participants in the hypothermic group were cooled by a forced-air cooling and heating blanket set to deliver unwarmed air (room temperature); a cooling and heating mattress connected to a heating-cooling unit set to 20.0 °C; and intravenous fluids and anaesthetic gases that were not warmed until the clip was applied. Participants in the normothermic group were maintained at a target oesophageal temperature of 36.5 °C by a cooling and heating mattress set to 38.0 °C; a forced-air cooling and heating blanket set to 36.0 °C to 38.0 °C; and intravenous fluids and inspired anaesthetic gases warmed to 37.0 °C. At the time of clip application, patients assigned to the hypothermia group were significantly colder than those assigned to the normothermia group, with a median temperature of 33.7 °C versus 36.6 °C, respectively ($P < 0.0001$).

Todd (Todd 2005) used a forced-air blanket connected to a heating-cooling unit to control the temperature for all patients. Hypothermia was induced after the start of anaesthesia and positioning of the patient, with a target temperature of 32.5 °C to 33.5 °C. The time from induction of anaesthesia to first clip placement was 217 ± 101 minutes and the temperature was 33.3 ± 0.8 °C. Rewarming of patients began after the last aneurysm clip had been secured and the temperature two hours after surgery was 36.4 ± 1.0 °C. The temperature of the normothermic group was kept between 36.0 °C and 37 °C and the time from induction of anaesthesia to the first clip placement was 214 ± 82 minutes. Both groups, normothermic and hypothermic, otherwise received similar treatment.

Els (Els 2006) induced hypothermia immediately after the operation with a target temperature of 35.0 °C. In the hypothermic group 10 participants were cooled by an intravenous cooling device; two participants were cooled by an external cooling device in combination with cool ventilator air fanning the body surface. All participants were cooled for 48 hours and active rewarming at a rate of 1 °C per 24 hours was applied. The normothermic group received the same treatment as the hypothermic group, although details were not provided regarding how the equality of ancillary treatment was ensured. The authors did not report how the temperature was maintained in the normothermic group.

Qiu (Qiu 2007) induced hypothermia for four days immediately after the operation, at a temperature of 33.0 °C to 35.0 °C, with the use of a water-circulating cooling blanket and, if necessary, a cooling cap around the head with circulating water at a temperature about 4 °C. The target temperature was generally achieved within 2.5 hours. For those who failed to achieve the target temperature within two hours refrigerated ice bags were placed around the neck. Patients were passively rewarmed over 10 to 24 hours. The normothermic group received the same treatment as the hypothermic group in all other ways, although again details were lacking as to how this was ensured. The authors did not report how the temperature was maintained in the normothermic group.

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Ongoing studies

One new ongoing study (Nuegebauer 2013) was identified by the updated search (Characteristics of ongoing studies). No new completed eligible studies were identified.

Excluded studies

A total of 29 publications were excluded after the review of the full reports. Three were non-randomized comparative trials (Bacher 1997; Beaufort 1995; Sato 2000); 13 studies presented an inappropriate comparator intervention (Baker 1994; Chouhan 2006; Gal 2002; Harris 2009; Hashiguchi 2003; Iwata 2006; Jian 2003; Karibe 2000; Leslie 2000; Li 2009; Liu 2006; Mrlian 2006; Shiozaki 2001); 12 were case series (Bernard 1999; Clifton 1992; Gal 2008; Gasser 2003; Iatrou 2002; Jian 2003; Kimme 2004; Maekawa 1994; Murthy 2005; Schwab 1998; Schwab 1998b; Shiozaki 2003) and one was a case control study (Himmelseher 2004). See Characteristics of excluded studies.

Risk of bias in included studies

All included trials have been described in the table Characteristics of included studies with potential bias being detailed in the accompanying 'Risk of bias' table (Figure 2; Figure 3).

Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

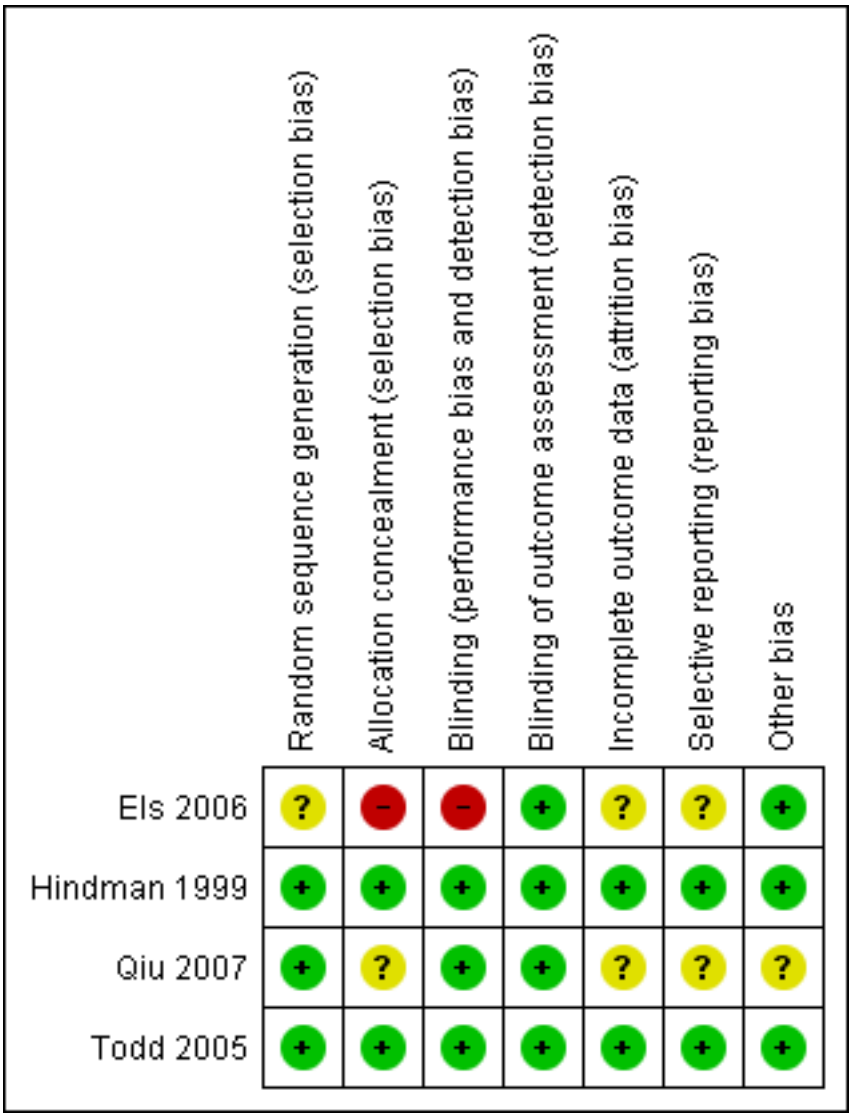

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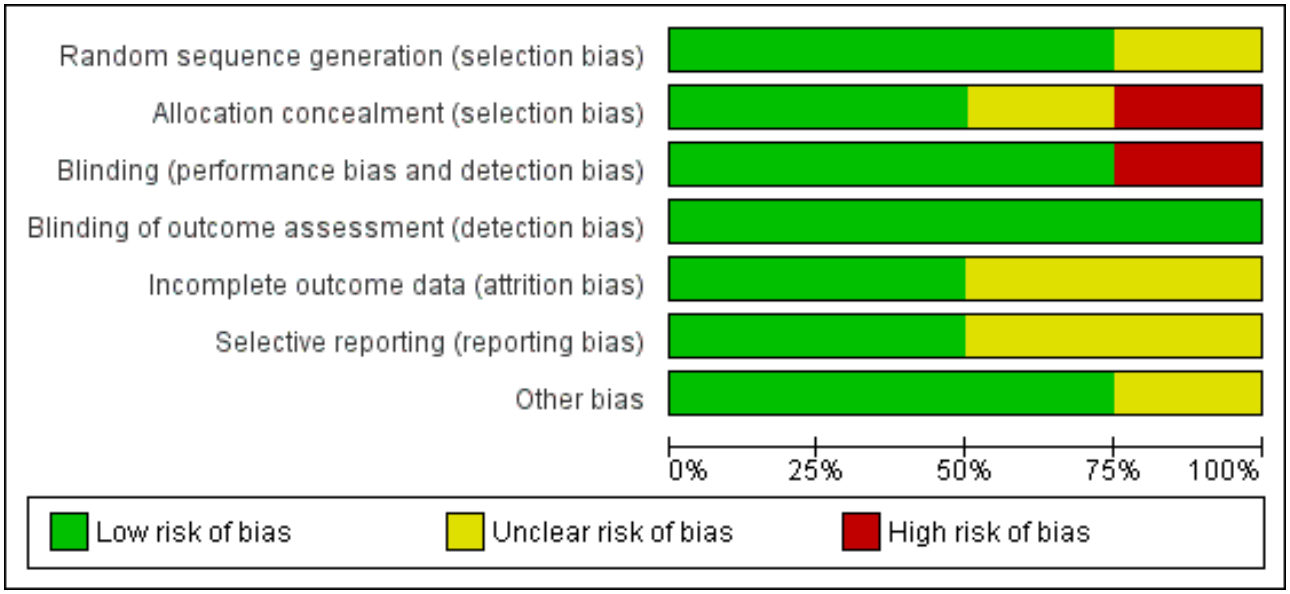
Figure 3. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.



Allocation

Three of the included studies (Hindman 1999; Qiu 2007; Todd 2005) provided details of the randomization process used. Methods used varied in rigour from a simple randomization table to permuted block randomization but, since a reasonable method was used in all three studies, they were each judged to have a low risk of selection bias due to inadequate randomization. One study (Els 2006) stated that the participants were randomly assigned to the interventions but provided no details of the method used and was therefore judged to be at unclear risk of bias in this domain.

Regarding allocation concealment, two studies described robust methods (Hindman 1999; Todd 2005), the third (Qiu 2007) mentioned that allocation concealment was used but did not describe a method. and the fourth (Els 2006) did not mention its use. The risk of bias in this domain was therefore judged to be low in the studies by Hindman and Todd (Hindman 1999; Todd 2005), unclear in the study by Qiu (Qiu 2007), and high in the study by Els (Els 2006).

Blinding

Two studies (Hindman 1999; Todd 2005) provided clear details regarding blinding of study personnel and outcome assessors and were assigned a low risk of bias in these domains. In the study by Qiu (Qiu 2007) the study personnel were unaware of treatment assignments but it was not clear if the outcome assessors were blinded. This may have had implications for more subjective outcomes including neurologic recovery. However, as the GOS clearly defined the scoring criteria it was unlikely that the lack of blinding of outcome assessors resulted in significant bias, therefore the risk was judged to be low.

The study by Els (Els 2006) did not describe any method of blinding either study personnel or outcome assessors. For the reasons detailed above, it was assigned a low risk of bias due to not blinding the outcome assessors. However, as lack of blinding of study personnel left the potential for intentional or unintentional differences in ancillary treatment it was assigned a high risk of bias in this domain.

Incomplete outcome data

In the studies by Hindman and Todd (Hindman 1999; Todd 2005) the numbers lost to follow-up were relatively small, three and one respectively. These studies were therefore judged to be at a low risk of attrition bias. Information was not available regarding completeness of follow-up in the studies by Els and Qiu (Els 2006; Qiu 2007) and therefore they were both judged to be at unclear risk of attrition bias.

Selective reporting

Two studies reported outcomes as pre-specified (Hindman 1999; Todd 2005) while the other two studies (Els 2006; Qiu 2007) did not pre-specify their outcome measures. Risk of reporting bias was therefore judged to be unclear in the latter two studies (Els 2006; Qiu 2007).

Other potential sources of bias

In the study by Qiu (Qiu 2007), the normothermic group had brain temperatures between 36.5 ºC and 37.5 ºC with rectal temperature even higher at 37.5 ºC to 38 ºC. Some of the normothermic group were therefore hyperthermic, which may have worsened their outcome making the intervention (hypothermia) seem more effective than it really was.

Effects of interventions

See: **Summary of findings for the main comparison** Cooling for cerebral protection during brain surgery

Primary outcomes

1. Mortality during treatment or the follow-up period (Comparison 1, Outcome 1.1)

See Analysis 1.1.

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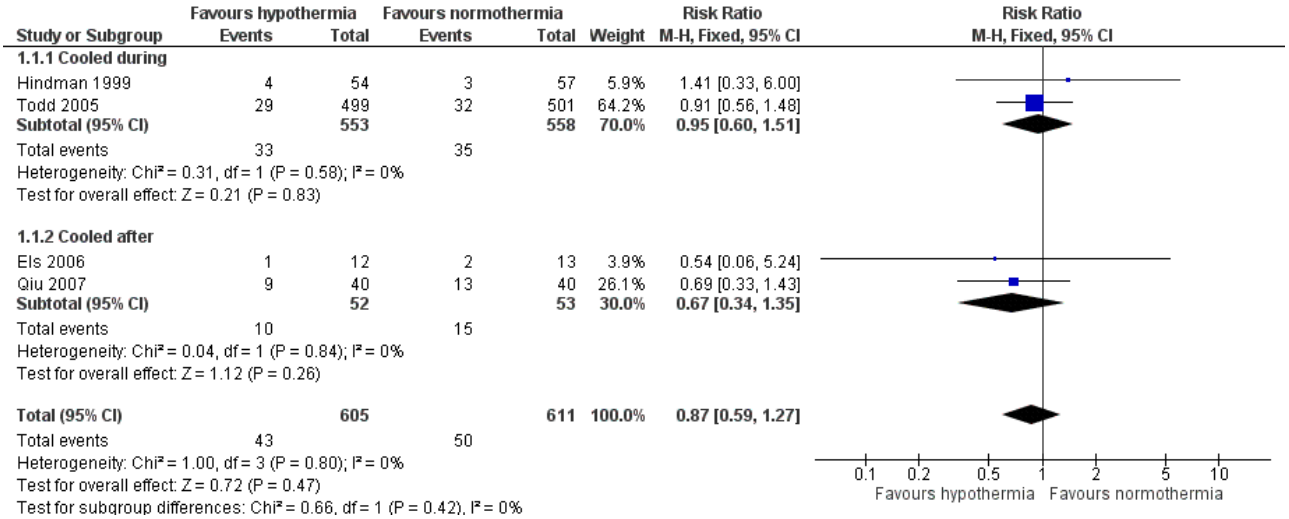
The mortality analysis involved 1216 of 1219 participants with three patients assigned to the hypothermic group being lost to final mortality follow-up and they were therefore excluded from the analysis for this outcome. Of these 1216 participants, 605 (49%) were allocated to hypothermia and 611 (50%) were allocated to normothermia. Overall, 93/1216 (7.6%) died during the treatment or the follow-up period, 43 (7.1%) of the hypothermic group and 50 (8.2%) of the normothermic group. This small difference was not statistically significant. The RR of dying if allocated to hypothermia compared to normothermia was 0.87 (95% CI 0.59 to 1.27, P = 0.47, I^2 = 0%). There was no indication that the time at which cooling was started affected the risk of dying (RR with intraoperative cooling 0.95, 95% CI 0.60 to 1.51, P = 0.83, I^2 = 0%; RR for cooling postoperatively 0.67, 95% CI 0.34 to 1.35, P = 0.26, I^2 = 0%) (Figure 4).

Figure 4. Forest plot of comparison: 1 Mortality, outcome: 1.1 Mortality rate.



The sensitivity analysis where studies at high risk of bias were removed (Els 2006) did not significantly change the estimate of the effect of hypothermia (RR 0.88, 95% CI 0.60 to 1.30, P = 0.53, I^2 = 0%) (Analysis 1.2). Three patients in the hypothermia arm were lost to follow-up as reported by Hindman 1999. If the best case scenario was assumed (all those assigned to receive hypothermia for whom mortality data were missing were assumed to have survived) this estimate did not change significantly (RR 0.86, 95% CI 0.59 to 1.27, P = 0.46, I^2 = 0%) (Analysis 1.3). If the worst case scenario was assumed (all those assigned to receive hypothermia for whom mortality data were missing were assumed to have died) this estimate did not change significantly either (RR 0.92, 95% CI 0.63 to 1.35, P = 0.68, I^2 = 0%) (Analysis 1.4).

2. Glasgow Outcome Score (GOS) of 3 or below (persistent vegetative state, severe disability or death as defined in the GOS) (Comparison 2, Outcome 2.1)

See Analysis 2.1.

The neurological outcome analysis involved 1216 of 1219 participants with three patients assigned to the hypothermic group being lost to final neurological outcome follow-up and therefore excluded from the analysis for this outcome. In total, 186 (15.3%) of 1216 participants had a GOS of three or less at follow-up assessment, 82 (13.5%) in the hypothermic group and 104 (17%) in the normothermic group. The risk of having a poor neurological outcome with a GOS of 3 or less if allocated to hypothermia versus normothermia was not statistically significant with a RR of 0.80 (95% CI 0.61 to 1.04, P = 0.09, I^2 = 0%) (Analysis 2.1). There was no indication that the time at which cooling was started affected this result (RR with intraoperative cooling 0.82, 95% CI 0.61 to 1.09, P = 0.17, I^2 = 0%; and RR for cooling postoperatively 0.67, 95% CI 0.34 to 1.35, P = 0.26, I^2 = 0%) (Figure 5).

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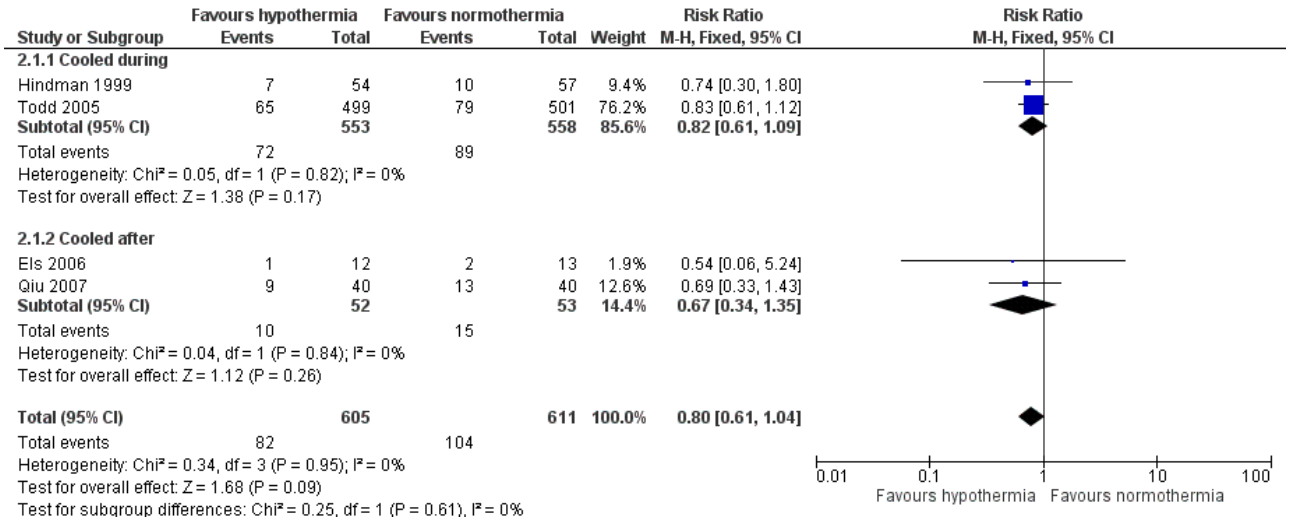
Figure 5. Forest plot of comparison: 2 Glasgow Outcome Score (GOS), outcome: 2.1 GOS.



Sensitivity analysis where studies judged to be at high risk of bias were removed (Els 2006) did not significantly alter this estimate (RR 0.80, 95% CI 0.61 to 1.05, P = 0.10, I^2 = 0%) (Analysis 2.2). Three participants in the hypothermia arm were lost to follow-up as reported by Hindman 1999. If the best case scenario was assumed (all those assigned to receive hypothermia for whom neurological outcome data were missing were assumed to have a good outcome with a GOS of 4 or more) this estimate did not change significantly (RR 0.79, 95% CI 0.61 to 1.03, P = 0.09, I^2 = 0%) (Analysis 2.3). If the worst case scenario was assumed (all those assigned to receive hypothermia for whom neurological outcome data were missing were assumed to have a poor outcome with a GOS of 3 or less), this estimate did not change significantly either (RR 0.82, 95% CI 0.63 to 1.07, P = 0.14, I^2 = 0%) (Analysis 2.4).

Secondary outcomes

3. Intracranial haemorrhage (Comparison 3)

Only Todd (Todd 2005) reported this outcome; 1000 participants (82% of the total in this review) were assessed. From these patients, 499 (49.9%) were allocated to hypothermia and 501 (50.1%) were allocated to normothermia. Overall, 85 (8.5%) developed an intracranial haemorrhage: 40 participants (8.0%) in the hypothermic group and 45 participants (9%) in the normothermic group. There was no statistically significant difference between these groups (RR 0.89, 95% CI 0.59 to 1.34, P = 0.58).

4. Ischaemic stroke (Comparison 4, Outcome 4.1)

See Analysis 4.1.

Hindman 1999 and Todd 2005 reported outcomes involving 1114 participants (91.3% of the total in this review); 556 (49.9%) were allocated to hypothermia and 558 (50%) were allocated to normothermia. Overall, 524 (47%) developed an ischaemic stroke during the treatment or the follow-up period, 252 (45.3%) of the hypothermic group and 272 (48.7%) of the normothermic group. This difference was not statistically significant and there was no significant statistical heterogeneity across these studies. The decrease of ischaemic stroke in patients allocated to hypothermia compared to normothermia (RR) was 0.93, 95% CI 0.82 to 1.05, P = 0.24, I^2 = 18%.

Surgical infection complications (Comparison 5, Outcome 5.1)

See Analysis 5.1.

Three trials (Hindman 1999 and Todd 2005, included in the subgroup 'cooled during surgery', and Qiu 2007 included in the subgroup 'cooled after surgery') reported this outcome. They involved 1194 participants (97.9% of the total in this review): 596 (49.9%) were allocated to hypothermia and 598 (50%) were allocated to normothermia. Overall, 385 (32.2%) developed surgical infection complications during the treatment or the follow-up period, 199 (33.4%) of the hypothermic group and 186 (31.1%) of the normothermic group. This difference was not statistically significant but there was significant statistical heterogeneity across these studies. The RR of surgical infection in participants allocated to hypothermia compared to normothermia was 1.20 (95% CI 0.73 to 1.97, P = 0.48, I^2 = 58%).

There was some indication that the time at which cooling was started may have affected the risk of infection. For the intraoperative cooling group there was no statistical significance (RR 1.03, 95% CI 0.87 to 1.22, P = 0.76, I^2 = 0%) but for those cooled postoperatively there was a statistically significant increase in the risk of infection in the hypothermic group (RR 1.77, 95% CI 1.05 to 2.98, P = 0.03).

6. Myocardial infarction (Comparison 6, Outcome 6.1)

See Analysis 6.1.

Three trials (Els 2006; Hindman 1999; Todd 2005) reported this outcome. They involved 1139 participants (93.4% of the total in this review): 568 (49.8%) were allocated to hypothermia and 571 (50.1%) were allocated to normothermia. Overall, 17 (1.5%) had a myocardial infarction during the treatment or the follow-up period, 11 (1.9%) of the hypothermic group and 6 (1%) of the normothermic group. This difference was not statistically significant and there was no indication of significant statistical heterogeneity across these studies. The RR of myocardial infarction in participants allocated to hypothermia compared to normothermia was 1.86 (95% CI 0.69 to 4.98, P = 0.22, I^2 = 0%). There was no indication that the time at which cooling was started affected the risk of myocardial infarction

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(RR with intraoperative cooling 2.01, 95% CI 0.69 to 5.83, P = 0.20; RR for cooling postoperatively 1.08, 95% CI 0.08 to 15.46, P = 0.95).

7. Congestive heart failure (Comparison 7, Outcome 7.1)

See Analysis 7.1.

Two trials (Hindman 1999; Todd 2005) reported this outcome. They involved 1114 participants (91.3% of the total in this review): 556 (49.9%) were allocated to hypothermia and 558 (50%) were allocated to normothermia. Overall, 111 (9.9%) had congestive heart failure during the treatment or the follow-up period, 51 (9.1%) of the hypothermic group and 60 (10.7%) of the normothermic group. This difference was not statistically significant and there was no indication of statistical heterogeneity across these studies. The RR of congestive heart failure for patients allocated to hypothermia compared to normothermia was 0.85 (95% CI 0.60 to 1.21, P = 0.38, I^2 = 0%).

8. Any other adverse events reported

None of the trials in this review reported any adverse effects directly related to the therapy other than those reported above.

DISCUSSION

Summary of main results

The effect estimates for the main outcomes are detailed in the summary of findings table (Summary of findings for the main comparison). Among patients undergoing craniotomy for various neurosurgical indications including aneurysm clipping, traumatic brain injury and Ischaemic stroke:

- there is no evidence that intraoperative or postoperative hypothermia significantly reduces or increases mortality;
- there is no evidence that intraoperative or postoperative hypothermia significant alters the risk of severe neurologic disability;
- there is no evidence that intraoperative hypothermia significantly alters the risk of intracranial haemorrhage, ischaemic stroke or congestive cardiac failure;
- there is no evidence that intraoperative or postoperative hypothermia significantly alters the risk of myocardial infarction;
- there is some weak evidence that postoperative hypothermia may increase the risk of infective complications.

Overall completeness and applicability of evidence

The findings of this review need to be interpreted in light of the following limitations.

Firstly, there is an overall lack of eligible high quality randomized controlled trials. The original search and this updated search identified only four eligible studies and of these only two (Hindman 1999; Todd 2005) were judged to have an overall low risk of bias. The lack of adequate evidence limited the accuracy and reliability of the results. For the same reason, the scope of the review was limited and it was not unable to fulfil its objectives in terms of analysing the effectiveness and safely of hypothermia by age group or underlying neurosurgical condition.

Secondly, there was potentially important clinical heterogeneity in terms of the underlying neurosurgical condition in the four eligible studies. While all patients in both the Hindman and Todd studies (Hindman 1999; Todd 2005) had surgery for aneurysm clipping, the study by Hindman (Hindman 1999) included both those who had and had not had a subarachnoid haemorrhage while the study by Todd (Todd 2005) was confined to those who had a confirmed subarachnoid haemorrhage. However, participants were homogenous in terms of premorbid neurologic disability, all being required to have a premorbid Modified Rankin score of 0 or 1 (Appendix 9), and patients in both trials who had had a subarachnoid haemorrhage were required to have a World Federation of Neurological Surgeons score of I, II or III (Appendix 8). The participants in the other two studies were more diverse as Els (Els 2006) studied those patients having decompressive craniectomy for cerebral oedema after acute ischaemic stoke and Qiu (Qiu 2007) studied those who had undergone craniotomy following traumatic brain injury. Among these two studies (Els 2006; Qiu 2007) there were differences in both the duration of cooling and ancillary treatment, and either of these may have contributed to the diversity and heterogeneity. Pooled estimates of effect that included those studies (Els 2006; Qiu 2007) in particular should therefore be interpreted with this limitation in mind.

Thirdly, over 90% of the participants underwent surgery for aneurysm clipping (Hindman 1999; Todd 2005), limiting the applicability of these results to other neurosurgical populations. The results of this review should be viewed at best as a general indication of the effectiveness and safety of intraoperative and postoperative hypothermia in patients undergoing craniotomy, with the important caveat that no robust or reliable conclusions can be drawn from it regarding the effects of hypothermia in specific neurosurgical populations aside perhaps from those undergoing cerebral aneurysm clipping.

Quality of the evidence

The quality of the evidence contributing to the main outcomes is detailed in the summary of findings table (Summary of findings for the main comparison).

The overall quality of the evidence contributing to the effect estimates for mortality and the Glasgow Outcome Score was deemed to be moderate, with data coming from two studies with an overall low risk of bias, a third with a low or unclear risk of bias (Qiu 2007) and the fourth having a high or unclear risk of bias (Els 2006).

For myocardial infarction, the quality of the evidence was rated as moderate as one of the studies had a high or unclear risk of bias (Els 2006).

The quality of the evidence contributing to effect estimates for ischaemic stoke and congestive heart failure was deemed to be high as both studies reporting these outcomes had a low risk of bias (Hindman 1999; Todd 2005).

For the estimate of infective complications, the quality of the evidence was judged to be low as there was significant statistical heterogeneity between studies and we lacked information on how infective complications were defined and diagnosed. The numbers analysed by individual studies for each infective complication were also small and the confidence intervals were not reported.

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Potential biases in the review process

An unrestricted comprehensive literature search was conducted for both the original review and the updated review with the aim of identifying all studies eligible for inclusion, and so to minimize reporting and selection bias. The small number of eligible studies that were identified, however, means that the effect estimates should not be taken as an accurate indication of the true effectiveness and safely of induced hypothermia across neurosurgical populations.

Agreements and disagreements with other studies or reviews

Two other recent systematic reviews addressing the effectiveness and safety of perioperative hypothermia in patients undergoing surgery for intracranial aneurysms found similar results (Li 2012; Zhao 2012). A Cochrane review (Li 2012) of mild intraoperative hypothermia in patients undergoing surgery for both ruptured and unruptured intracranial aneurysms did not show any statistically significant difference in outcomes for those treated with hypothermia versus normothermia. A second systematic review (Zhao 2012) addressing the use of intraoperative hypothermia in patients undergoing intracranial aneurysm surgery similarly did not show any significant differences in mortality, neurologic outcome or adverse events between the groups. Like this review, both were limited by the lack of availability of eligible studies and both concluded that currently there was insufficient evidence to support the routine use of hypothermia in this population.

We found only one eligible study (Qiu 2007) addressing the role of hypothermia in patients undergoing surgery following traumatic brain injury, and it is important to note that. That study had a number of methodological problems limiting the ability of this review to determine the role of hypothermia in this population. A post hoc analysis (Clifton 2012) of two randomized controlled trials of hypothermia for severe traumatic brain injury looked at the effect of hypothermia in those patients who had haematomas surgically evacuated within 24 hours of injury. They found that induction of hypothermia to 35 °C before or soon after craniotomy and maintenance at 33 °C for 48 hours postoperatively was associated with an improved neurologic outcome at six months. They postulated that early induction of hypothermia may be beneficial in those requiring surgical evacuation of post-traumatic intracranial haematoma and emphasize the need for future trials in this area that target similar temperature goals.

There is a paucity of trials addressing the value of hypothermia in all neurosurgical populations, especially in those undergoing surgery for epilepsy, non-aneurysmal vascular deformities and ischaemic stoke. Further evidence is required in all these groups before the true effects of hypothermia can be determined in these populations.

AUTHORS' CONCLUSIONS

Implications for practice

At present there is no evidence to support the use of hypothermia in patients undergoing craniotomy, although further high quality studies are required to determine its true efficacy and safety across different neurosurgical populations.

Implications for research

More clinical trials are needed to establish the effectiveness of hypothermia for cerebral protection during brain surgery before drawing any firm conclusions with regards to its potential in these patients. There is a need for more well designed and conducted clinical trials to detect clinically important differences between hypothermia and normothermia. More information is also needed on the minimum effective target temperature, the time period during which cooling is appropriate, the best rewarming strategies and potential complications.

ACKNOWLEDGEMENTS

We wish to acknowledge WRO Milani, PL Antibas and GF Prado, the authors of the original review (Milani 2011) on this topic. Although they did not directly participate in this updated version, all made a substantial and significant contribution to this review by conceiving this topic and building the foundations upon which this updated version was constructed.

We would like to thank Jane Cracknell and Karen Hovhannisyan of The Cochrane Collaboration for their assistance during the preparation of this review. We also wish to thank Professor Micheal Bennett for his editorial expertise and advice.

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Els 2006

Methods	Randomized controlled trial conducted between June 2000 and June 2003
Participants	A total of 25 participants with a mean age of 49 years **Inclusion criteria** • typical clinical signs of a larger supratentorial infarction • signal changes in early diffusion - weighted and perfusion-weighted MRI • age < 65 years • no pre-existing disability • no dementia • no tegmental syndrome **Exclusion criteria** • not implicitly stated
Interventions	Hemicraniectomy alone (n = 13) versus hemicraniectomy and hypothermia (35 °C) (n = 12)

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Els 2006 *(Continued)*

	Hypothermia was induced immediately postoperatively and achieved using an intravenous cooling device in 10 participants and an external cooling device in 2 participants. Cooling was maintained for 48 hours
Outcomes	Duration of ICU admission • Duration of mechanical ventilation • Duration and maximal dosage of catecholamine treatment • Mortality due to hypothermia related complications • Mortality due to perioperative complications
Notes	• No details regarding how randomization was performed • No sample size calculation reported • Efforts were made to ensure equality of ancillary treatment between the groups but details are lacking as to the effectiveness of these efforts • Intention-to-treat analysis not used

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	The authors report that the participants were randomized to one of two treatment arms but the method used is not reported
Allocation concealment (selection bias)	High risk	No method of allocation concealment is described
Blinding (performance bias and detection bias) All outcomes	High risk	No method of blinding study personnel is described
Blinding of outcome assessment (detection bias) All outcomes	Low risk	It is not clear if those assessing outcomes were blinded to treatment assignment but as outcomes were largely objective the risk of bias was judged to be low
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Not enough information available to determine completeness of follow-up
Selective reporting (reporting bias)	Unclear risk	Outcomes and their relative priorities were not clearly pre-specified making it difficult to determine if they were reported as intended
Other bias	Low risk	None

Hindman 1999

Methods	Mulitcentre randomized controlled trial conducted between October 1994 and July 1996
Participants	A total of 114 participants with a median age of 52 years **Inclusion criteria** • Patients with intracranial aneurysms who had a pre-hospitalization or pre-SAH Rankin disability score of 0 or 1 • Patients with acute SAH with a World Federation of Neurosurgical Surgeons score of I, II or III

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Hindman 1999 *(Continued)*

	Exclusion criteria • Intubated at the time of enrolment • Cryoglobulinaemia • Severe Raynuad's disease • Sickle cell disease • Another disorder considered to be a contraindication to hypothermia • Entered into a blinded trial of any drug which may affect neurologic outcome
Interventions	Normothermia (36.5 °C) versus hypothermia (33.5 °C). Temperature targets were achieved for both groups using cooling-heating mattresses and forced-air devices in addition to manipulation of the temperature of intravenous fluids and anaesthetic gases. Cooling or heating was commenced after induction of anaesthesia and positioning of the patient and before surgical incision. Those assigned to the hypothermia group were cooled until the time of clip application, after which they were actively rewarmed
Outcomes	• Mortality (in-hospital and at 3 to 6 months) • National Institute of Health Care Stroke Scale • Glasgow Outcome Score • Delayed Ischaemic deficit • Duration of hospital admission • Requirement for invasive ventilation • Requirement for intensive care • Postoperative pneumonia • Postoperative cardiovascular complications • Postoperative hydrocephalus
Notes	Pilot study, no sample size reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomization was done by sequentially numbered envelopes
Allocation concealment (selection bias)	Low risk	Envelopes were prepared by the co-ordinating centre, University of Iowa and opened by the anaesthesiologist at the time of start of study treatment
Blinding (performance bias and detection bias) All outcomes	Low risk	Adequate. Group assignment was not revealed to the surgical team
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Adequate. Group assignment was not revealed to those assessing outcomes
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only three patients were lost to follow-up
Selective reporting (reporting bias)	Low risk	Outcomes were reported as pre-specified
Other bias	Low risk	None

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Qiu 2007

Methods	Randomized controlled trial conducted between January 2002 and December 2003
Participants	A total of 80 participants with mean of age of 41 years in the hypothermic group and 40 years in the normothermic group **Inclusion criteria** • History of traumatic brain injury • Brain injury confirmed by sequential CT scanning within 6 hours after trauma • Glasgow Coma Scale scores of 8 or less **Exclusion criteria** • Pregnancy • Age < 19 years or > 65 years • Multiple trauma • Haemorrhagic shock • Absent brain stem reflex on initial examination • Previous disabling neurologic disease
Interventions	Normothermic (unstated target temperature but actual brain temperature was 36.6 °C to 37.5 °C) versus hypothermic (33 °C to 35 °C) group. For those assigned to the hypothermia group cooling was commenced after craniotomy and maintained for 4 days postoperatively after which they were passively rewarmed
Outcomes	• Intracranial pressure • Serum superoxide levels • Glasgow Outcome Score at one year • Adverse events including pulmonary infections, thrombocytopenia, haemorrhage, hypoglycaemia etc • Mortality at 1 year
Notes	• No sample size reported • Rectal temperature reached as high as 38 °C in the normothermia group, which may account in part for the favourable neurologic outcome observed in the hypothermia group • Efforts were made to ensure equality of ancillary treatment between the groups but there is no report of how effective these efforts were • No report of completeness of or losses to follow-up

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A randomization table was used
Allocation concealment (selection bias)	Unclear risk	The authors state that allocation concealment was used but do not describe how it was achieved
Blinding (performance bias and detection bias) All outcomes	Low risk	Physicians were not involved in data collection; nursing staff and the surgical team were not aware of the patient's group allocation

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Qiu 2007 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	Low risk	It is not clear if those assessing outcomes were blinded to treatment assignment but as outcomes were largely objective the risk of bias was judged to be low
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No details are provided regarding completeness of follow-up
Selective reporting (reporting bias)	Unclear risk	Outcomes are not clearly pre-specified so it is difficult to exclude selective reporting
Other bias	Unclear risk	Rectal temperature reached as high as 38 °C in the normothermia group, which may account in part for the favourable neurologic outcome observed in the hypothermia group

Todd 2005

Methods	Multicentre randomized controlled trial conducted from February 2000 to April 2003
Participants	A total of 1000 participants with a mean age of 52 years **Inclusion criteria** • Age 18 years or more • Subarachnoid haemorrhage with a radiologically demonstrated intracranial aneurysm within 14 days of surgery • World Federation of Neuroligical Surgeons score of I, II or III • Rankin Score 0 or 1 before haemorrhage **Exclusion criteria** • Body mass index > 35 • Cold related disorder • Endotracheal tube in situ
Interventions	Normothermia (36 °C to 37 °C) versus hypothermia (32.5 °C to 33.5 °C). Warming or cooling was commenced after induction of anaesthesia with the goal of achieving target temperature by the time the first aneurysm clip was applied. Rewarming began after the last clip was in place
Outcomes	• Glasgow Outcome Score • Rankin score • Barthel index • National Institutes of Health Stroke Scale • Neuropyschological assessment • Duration of intensive care and hospital admission • Mortality at 90 days • Adverse events including neurological, cardiovascular, haematological and infective complications
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement

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Todd 2005 *(Continued)*

Random sequence generation (selection bias)	Low risk	Permuted block scheme with stratification according to centre and interval between subarachnoid haemorrhage and time of surgery
Allocation concealment (selection bias)	Low risk	Allocation assignment was by means of a telephone assessed computerized system which directed the anaesthesiologist to open an opaque envelope containing the individual patient's treatment assignment
Blinding (performance bias and detection bias) All outcomes	Low risk	All study personnel except the anaesthesiologist were unaware of the individual patient treatment assignment
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Outcome assessors were unaware of treatment assignments
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only one patient was lost to follow-up. This patient was not included in their final analysis, which was based on a total of 1000 patients
Selective reporting (reporting bias)	Low risk	None
Other bias	Low risk	None

ICU = intensive care unit
GCS = Glasgow Coma Scale
SAH = acute aneurysmal subarachnoid haemorrhage
ICP = intracranial pressure

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Bacher 1997	Controlled trial evaluating metabolic rather than clinical outcomes
Baker 1994	Randomized controlled trial evaluating physiological rather than clinical outcomes
Beaufort 1995	Controlled trial evaluating effects of hypothermia on duration of action of muscle relaxant medication
Bernard 1999	Case series
Chouhan 2006	Randomized controlled trial, however due to equipment limitations there was considerable delay in achieving target in 32% of patients assigned to the hypothermia group and failure to maintain normothermia in 34% of patients assigned to that treatment arm
Clifton 1992	Case series
Gal 2002	Small quasi-randomized trial of hypothermia for traumatic brain injury in which several of the patients assigned to the hypothermic group did not undergo neurosurgery
Gal 2008	Case series
Gasser 2003	Case series

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Study	Reason for exclusion
Harris 2009	Randomized controlled trial of hypothermia for traumatic brain injury in which not all participants underwent surgery
Hashiguchi 2003	Randomized controlled trial of hypothermia for traumatic brain injury in which not all participants underwent surgery
Himmelseher 2004	Case-control study
Hindman 2009	Post hoc analysis of previous randomized controlled trial
Iatrou 2002	Case series
Iwata 2006	Randomized controlled trial evaluating non-clinical outcomes
Jian 2003	Case series
Jiang 2006	Randomized controlled trial comparing two hypothermic groups
Karibe 2000	Randomized controlled trial evaluating non-clinical outcomes
Kimme 2004	Case series
Leslie 2000	Randomized controlled trial, however the authors did not evaluate any of the outcomes proposed by this review
Li 2009	Randomized controlled trial of hypothermia for traumatic brain injury in which not all participants underwent surgery
Liu 2006	Randomized controlled trial of hypothermia for traumatic brain injury in which not all participants underwent surgery
Maekawa 1994	Case series
Mrlian 2006	Randomized controlled trial of hypothermia for traumatic brain injury in which not all participants underwent surgery
Murthy 2005	Case series
Nguyen 2010	Post hoc analysis of previous randomized controlled trial
Sato 2000	Controlled trial evaluating non-clinical outcomes
Schwab 1998	Case series
Schwab 1998b	Case series
Shiozaki 2001	Randomized controlled trial of hypothermia for traumatic brain injury in which not all participants underwent surgery
Shiozaki 2003	Case series

Characteristics of ongoing studies *[ordered by study ID]*

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Nuegebauer 2013

Trial name or title	DEcompressive surgery Plus hypoTHermia for Space-Occupying Stroke (DEPTH-SOS): a protocol of a multicenter randomized controlled clinical trial and a literature review
Methods	Two arm multicentre randomized controlled trial
Participants	Patients with malignant middle cerebral artery infarction
Interventions	Decompressive craniectomy together with induced hypothermia
Outcomes	Primary outcome: mortality at day 14. Secondary outcomes: functional outcome at day 14 and at 12 months and complications related to hypothermia
Starting date	10 August 2011
Contact information	Eric Jüttler, e-mail: eric.juettler@uni-ulm.de
Notes	Currently recruiting. Target sample size 50

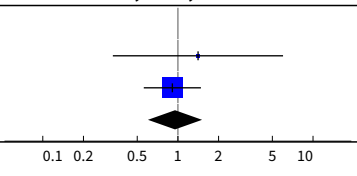
DATA AND ANALYSES

Comparison 1. Mortality

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Mortality rate	4	1216	Risk Ratio (M-H, Fixed, 95% CI)	0.87 [0.59, 1.27]
1.1 Cooled during	2	1111	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.60, 1.51]
1.2 Cooled after	2	105	Risk Ratio (M-H, Fixed, 95% CI)	0.67 [0.34, 1.35]
2 Mortality rate (methodological quality)	3	1191	Risk Ratio (M-H, Fixed, 95% CI)	0.88 [0.60, 1.30]
3 Mortality rate (best case scenario)	4	1219	Risk Ratio (M-H, Fixed, 95% CI)	0.86 [0.59, 1.27]
4 Mortality rate (worst case scenario)	4	1219	Risk Ratio (M-H, Fixed, 95% CI)	0.92 [0.63, 1.35]

Analysis 1.1. Comparison 1 Mortality, Outcome 1 Mortality rate.

Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
1.1.1 Cooled during					
Hindman 1999	4/54	3/57		5.86%	1.41[0.33,6]
Todd 2005	29/499	32/501		64.16%	0.91[0.56,1.48]
Subtotal (95% CI)	**553**	**558**		**70.03%**	**0.95[0.6,1.51]**



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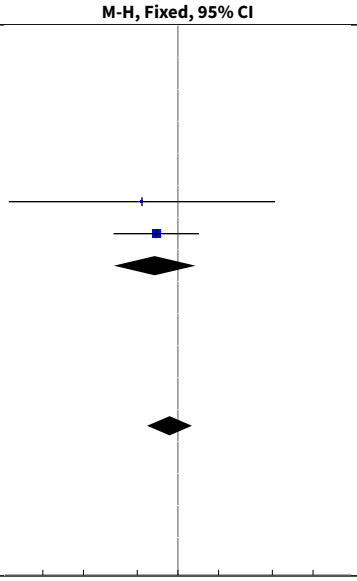
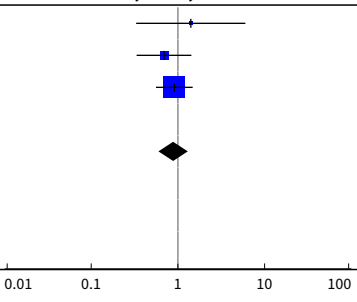
Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Total events: 33 (Favours hypothermia), 35 (Favours normothermia)					
Heterogeneity: Tau²=0; Chi²=0.31, df=1(P=0.58); I²=0%					
Test for overall effect: Z=0.21(P=0.83)					
1.1.2 Cooled after					
Els 2006	1/12	2/13		3.86%	0.54[0.06,5.24]
Qiu 2007	9/40	13/40		26.12%	0.69[0.33,1.43]
Subtotal (95% CI)	**52**	**53**		**29.97%**	**0.67[0.34,1.35]**
Total events: 10 (Favours hypothermia), 15 (Favours normothermia)					
Heterogeneity: Tau²=0; Chi²=0.04, df=1(P=0.84); I²=0%					
Test for overall effect: Z=1.12(P=0.26)					
Total (95% CI)	**605**	**611**		**100%**	**0.87[0.59,1.27]**
Total events: 43 (Favours hypothermia), 50 (Favours normothermia)					
Heterogeneity: Tau²=0; Chi²=1, df=3(P=0.8); I²=0%					
Test for overall effect: Z=0.72(P=0.47)					
Test for subgroup differences: Chi²=0.66, df=1 (P=0.42), I²=0%					



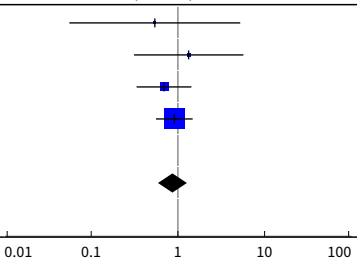
Analysis 1.2. Comparison 1 Mortality, Outcome 2 Mortality rate (methodological quality).

Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Hindman 1999	4/54	3/57		6.1%	1.41[0.33,6]
Qiu 2007	9/40	13/40		27.17%	0.69[0.33,1.43]
Todd 2005	29/499	32/501		66.74%	0.91[0.56,1.48]
Total (95% CI)	**593**	**598**		**100%**	**0.88[0.6,1.3]**
Total events: 42 (Favours hypothermia), 48 (Favours normothermia)					
Heterogeneity: Tau²=0; Chi²=0.84, df=2(P=0.66); I²=0%					
Test for overall effect: Z=0.64(P=0.53)					



Analysis 1.3. Comparison 1 Mortality, Outcome 3 Mortality rate (best case scenario).

Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Els 2006	1/12	2/13		3.85%	0.54[0.06,5.24]
Hindman 1999	4/57	3/57		6.02%	1.33[0.31,5.69]
Qiu 2007	9/40	13/40		26.08%	0.69[0.33,1.43]
Todd 2005	29/499	32/501		64.06%	0.91[0.56,1.48]
Total (95% CI)	**608**	**611**		**100%**	**0.86[0.59,1.27]**
Total events: 43 (Favours hypothermia), 50 (Favours normothermia)					



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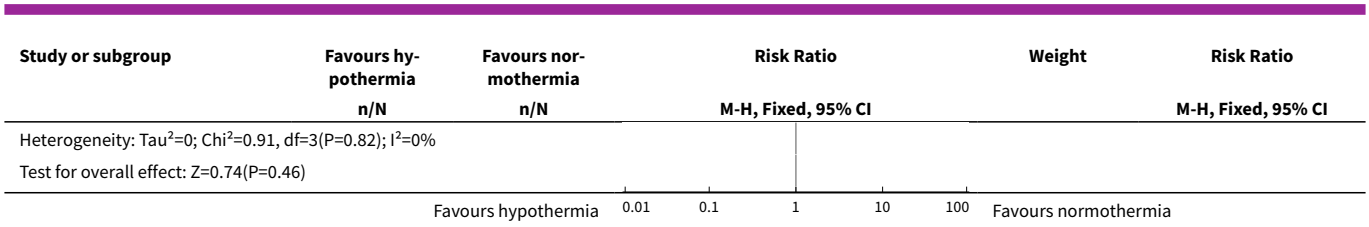
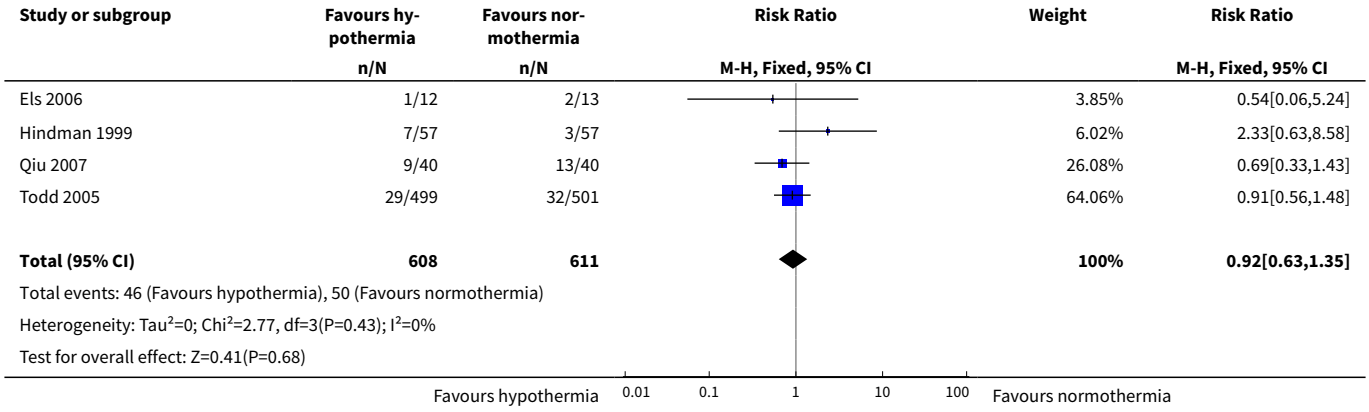


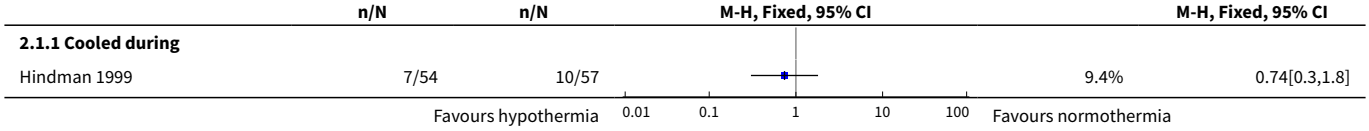
Analysis 1.4. Comparison 1 Mortality, Outcome 4 Mortality rate (worst case scenario).



Comparison 2. Glasgow Outcome Score (GOS)

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 GOS Score	4	1216	Risk Ratio (M-H, Fixed, 95% CI)	0.80 [0.61, 1.04]
1.1 Cooled during	2	1111	Risk Ratio (M-H, Fixed, 95% CI)	0.82 [0.61, 1.09]
1.2 Cooled after	2	105	Risk Ratio (M-H, Fixed, 95% CI)	0.67 [0.34, 1.35]
2 GOS Score (methodological quality)	3	1191	Risk Ratio (M-H, Fixed, 95% CI)	0.80 [0.61, 1.05]
3 GOS Score (best case scenario)	4	1219	Risk Ratio (M-H, Fixed, 95% CI)	0.79 [0.61, 1.03]
4 GOS Score (worst case scenario)	4	1219	Risk Ratio (M-H, Fixed, 95% CI)	0.82 [0.63, 1.07]

Analysis 2.1. Comparison 2 Glasgow Outcome Score (GOS), Outcome 1 GOS Score.



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Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Todd 2005	65/499	79/501		76.18%	0.83[0.61,1.12]
Subtotal (95% CI)	**553**	**558**		**85.58%**	**0.82[0.61,1.09]**
Total events: 72 (Favours hypothermia), 89 (Favours normothermia)					
Heterogeneity: Tau²=0; Chi²=0.05, df=1(P=0.82); I²=0%					
Test for overall effect: Z=1.38(P=0.17)					
2.1.2 Cooled after					
Els 2006	1/12	2/13		1.86%	0.54[0.06,5.24]
Qiu 2007	9/40	13/40		12.56%	0.69[0.33,1.43]
Subtotal (95% CI)	**52**	**53**		**14.42%**	**0.67[0.34,1.35]**
Total events: 10 (Favours hypothermia), 15 (Favours normothermia)					
Heterogeneity: Tau²=0; Chi²=0.04, df=1(P=0.84); I²=0%					
Test for overall effect: Z=1.12(P=0.26)					
Total (95% CI)	**605**	**611**		**100%**	**0.8[0.61,1.04]**
Total events: 82 (Favours hypothermia), 104 (Favours normothermia)					
Heterogeneity: Tau²=0; Chi²=0.34, df=3(P=0.95); I²=0%					
Test for overall effect: Z=1.68(P=0.09)					
Test for subgroup differences: Chi²=0.25, df=1 (P=0.61), I²=0%					

Analysis 2.2. Comparison 2 Glasgow Outcome Score (GOS), Outcome 2 GOS Score (methodological quality).

Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Hindman 1999	7/54	10/57		9.58%	0.74[0.3,1.8]
Qiu 2007	9/40	13/40		12.8%	0.69[0.33,1.43]
Todd 2005	65/499	79/501		77.62%	0.83[0.61,1.12]
Total (95% CI)	**593**	**598**		**100%**	**0.8[0.61,1.05]**
Total events: 81 (Favours hypothermia), 102 (Favours normothermia)					
Heterogeneity: Tau²=0; Chi²=0.22, df=2(P=0.89); I²=0%					
Test for overall effect: Z=1.63(P=0.1)					

Analysis 2.3. Comparison 2 Glasgow Outcome Score (GOS), Outcome 3 GOS Score (best case scenario).

Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Els 2006	1/12	2/13		1.85%	0.54[0.06,5.24]
Hindman 1999	7/57	10/57		9.64%	0.7[0.29,1.71]
Qiu 2007	9/40	13/40		12.53%	0.69[0.33,1.43]
Todd 2005	65/499	79/501		75.98%	0.83[0.61,1.12]

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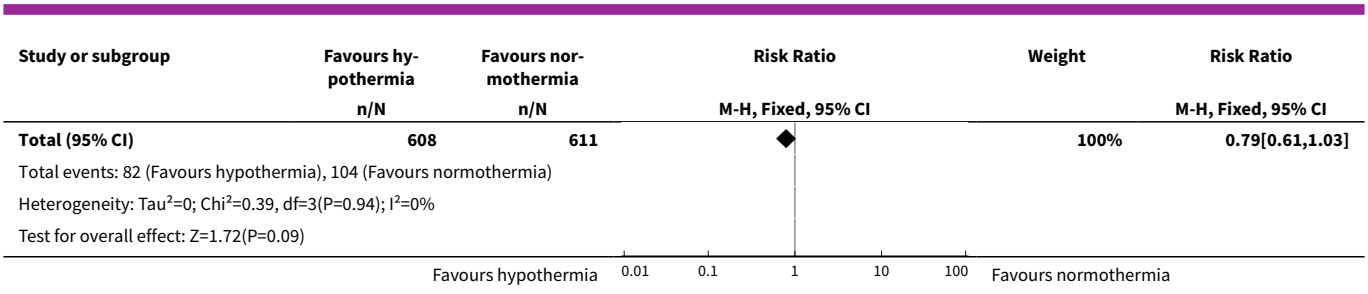
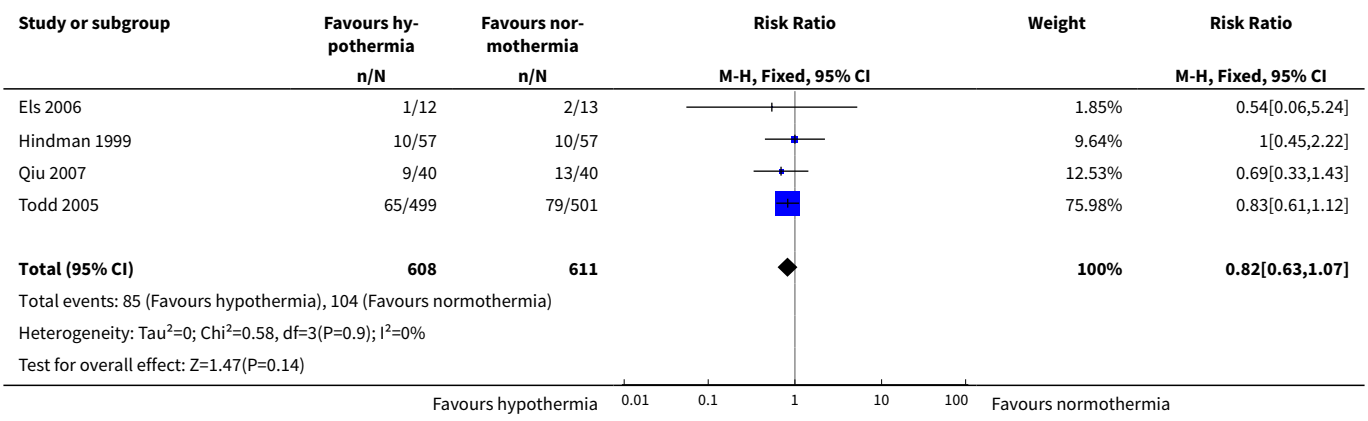


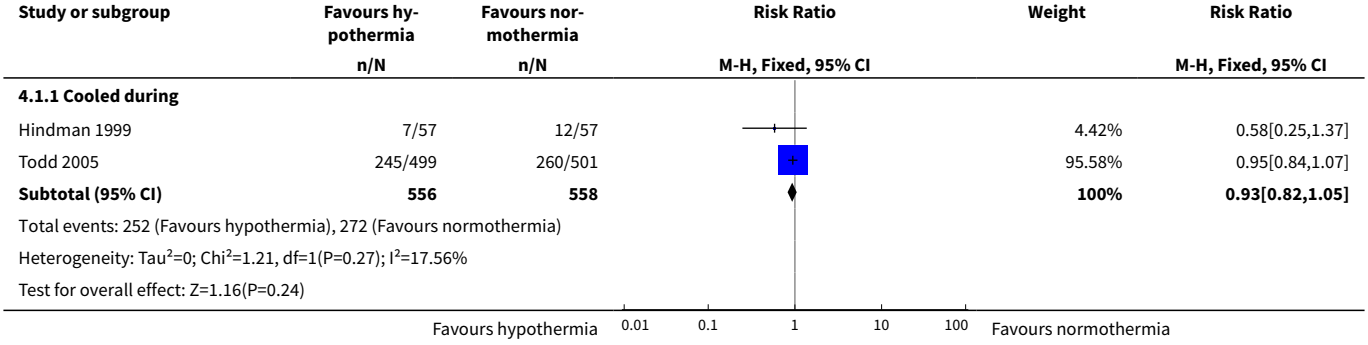
Analysis 2.4. Comparison 2 Glasgow Outcome Score (GOS), Outcome 4 GOS Score (worst case scenario).



Comparison 4. Ischaemic stroke

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Ischaemic stroke	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
1.1 Cooled during	2	1114	Risk Ratio (M-H, Fixed, 95% CI)	0.93 [0.82, 1.05]

Analysis 4.1. Comparison 4 Ischaemic stroke, Outcome 1 Ischaemic stroke.



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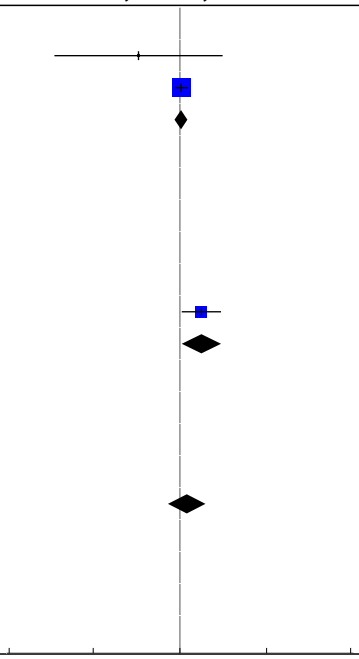
Comparison 5. Surgical infection complications

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Infections	3	1194	Risk Ratio (M-H, Random, 95% CI)	1.20 [0.73, 1.97]
1.1 Cooled during	2	1114	Risk Ratio (M-H, Random, 95% CI)	1.03 [0.87, 1.22]
1.2 Cooled after	1	80	Risk Ratio (M-H, Random, 95% CI)	1.77 [1.05, 2.98]

Analysis 5.1. Comparison 5 Surgical infection complications, Outcome 1 Infections.

Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Random, 95% CI	Weight	Risk Ratio M-H, Random, 95% CI
5.1.1 Cooled during					
Hindman 1999	1/57	3/57		4.6%	0.33[0.04,3.11]
Todd 2005	175/499	170/501		58.24%	1.03[0.87,1.23]
Subtotal (95% CI)	**556**	**558**		**62.84%**	**1.03[0.87,1.22]**
Total events: 176 (Favours hypothermia), 173 (Favours normothermia)					
Heterogeneity: Tau^2=0; Chi^2=0.98, df=1(P=0.32); I^2=0%					
Test for overall effect: Z=0.3(P=0.76)					
5.1.2 Cooled after					
Qiu 2007	23/40	13/40		37.16%	1.77[1.05,2.98]
Subtotal (95% CI)	**40**	**40**		**37.16%**	**1.77[1.05,2.98]**
Total events: 23 (Favours hypothermia), 13 (Favours normothermia)					
Heterogeneity: Not applicable					
Test for overall effect: Z=2.15(P=0.03)					
Total (95% CI)	**596**	**598**		**100%**	**1.2[0.73,1.97]**
Total events: 199 (Favours hypothermia), 186 (Favours normothermia)					
Heterogeneity: Tau^2=0.1; Chi^2=4.79, df=2(P=0.09); I^2=58.23%					
Test for overall effect: Z=0.71(P=0.48)					
Test for subgroup differences: Chi^2=3.8, df=1 (P=0.05), I^2=73.67%					



Comparison 6. Myocardial infarction

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Myocardial infarction	3	1139	Risk Ratio (M-H, Fixed, 95% CI)	1.86 [0.69, 4.98]
1.1 Cooled during	2	1114	Risk Ratio (M-H, Fixed, 95% CI)	2.01 [0.69, 5.83]
1.2 Cooled after	1	25	Risk Ratio (M-H, Fixed, 95% CI)	1.08 [0.08, 15.46]

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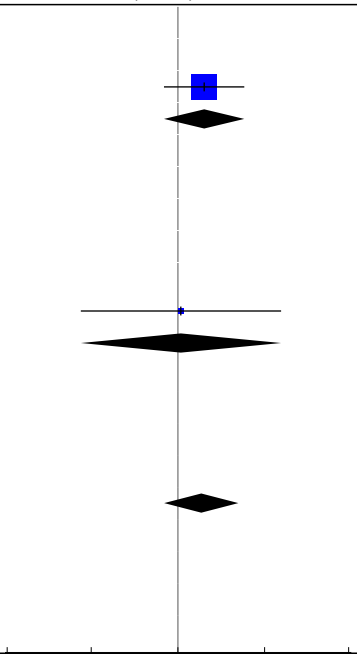
Analysis 6.1. Comparison 6 Myocardial infarction, Outcome 1 Myocardial infarction.

Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
6.1.1 Cooled during					
Hindman 1999	0/57	0/57			Not estimable
Todd 2005	10/499	5/501		83.87%	2.01[0.69,5.83]
Subtotal (95% CI)	**556**	**558**		**83.87%**	**2.01[0.69,5.83]**
Total events: 10 (Favours hypothermia), 5 (Favours normothermia)					
Heterogeneity: Not applicable					
Test for overall effect: Z=1.28(P=0.2)					
6.1.2 Cooled after					
Els 2006	1/12	1/13		16.13%	1.08[0.08,15.46]
Subtotal (95% CI)	**12**	**13**		**16.13%**	**1.08[0.08,15.46]**
Total events: 1 (Favours hypothermia), 1 (Favours normothermia)					
Heterogeneity: Not applicable					
Test for overall effect: Z=0.06(P=0.95)					
Total (95% CI)	**568**	**571**		**100%**	**1.86[0.69,4.98]**
Total events: 11 (Favours hypothermia), 6 (Favours normothermia)					
Heterogeneity: Tau^2=0; Chi^2=0.18, df=1(P=0.67); I^2=0%					
Test for overall effect: Z=1.23(P=0.22)					
Test for subgroup differences: Chi^2=0.18, df=1 (P=0.67), I^2=0%					

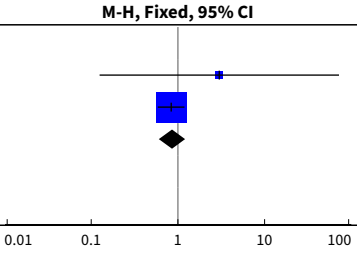


Comparison 7. Congestive heart failure

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Congestive heart failure	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
1.1 Cooled during	2	1114	Risk Ratio (M-H, Fixed, 95% CI)	0.85 [0.60, 1.21]

Analysis 7.1. Comparison 7 Congestive heart failure, Outcome 1 Congestive heart failure.

Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
7.1.1 Cooled during					
Hindman 1999	1/57	0/57		0.83%	3[0.12,72.13]
Todd 2005	50/499	60/501		99.17%	0.84[0.59,1.19]
Subtotal (95% CI)	**556**	**558**		**100%**	**0.85[0.6,1.21]**
Total events: 51 (Favours hypothermia), 60 (Favours normothermia)					
Heterogeneity: Tau^2=0; Chi^2=0.61, df=1(P=0.43); I^2=0%					



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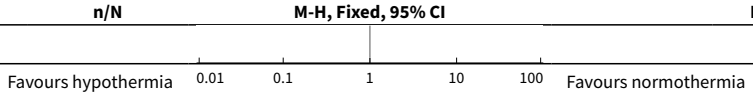
Study or subgroup	Favours hypothermia n/N	Favours normothermia n/N	Risk Ratio M-H, Fixed, 95% CI	Weight	Risk Ratio M-H, Fixed, 95% CI
Test for overall effect: Z=0.88(P=0.38)					



APPENDICES

Appendix 1. Glasgow Outcome Score

Score	Classification	Description
1	Death	-------
2	Persistent vegetative State	Patients who remain unresponsive and speechless for weeks or months until death after acute brain damage.
3	Severe disability	Conscious but disabled. Patients who are dependent for daily support by reason of mental or physical disability or by a combination of both.
4	Moderate disability	Disabled but independent. Patients can perform daily tasks of living. Employment is possible but may require special provisions.
5	Good recovery	Normal or near normal recovery.

Appendix 2. Search Strategy for the Cochrane Central Register of Controlled Trials (CENTRAL)

#1 MeSH descriptor Hypothermia, Induced explode all trees
#2 MeSH descriptor Hypothermia explode all trees
#3 MeSH descriptor Cryotherapy explode all trees
#4 MeSH descriptor Cryosurgery explode all trees
#5 Cooling or Cryotherapy or Hypotherm*
#6 hypotherm* near induced
#7 cold*
#8 cryo*
#9 (#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #8)
#10 MeSH descriptor Neurosurgical Procedures explode all trees
#11 MeSH descriptor Neurosurgery explode all trees
#12 MeSH descriptor Brain Injuries explode all trees
#13 MeSH descriptor Intracranial Aneurysm explode all trees
#14 MeSH descriptor Intracranial Arterial Diseases explode all trees
#15 MeSH descriptor Cerebrovascular Accident explode all trees
#16 MeSH descriptor Cerebrovascular Disorders explode all trees
#17 (operat* or surg* or procedur*) near (brain* or neuro*)
#18 Intracranial
#19 (Brain or Cerebro*) near (disord* or trauma* or damage)
#20 (#10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19)
#21 (#9 AND #20)

Appendix 3. Search strategy for MEDLINE (OvidSP)

1. exp Hypothermia Induced/ or Circulatory Arrest Deep Hypothermia Induced/ or exp Hypothermia/ or exp Cryotherapy/ or exp Cryosurgery/ or (Cooling or Cryotherapy or Hypotherm*).mp. or (cryo* or cold).ti,ab.

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2. exp Neurosurgical Procedures/ or exp Neurosurgery/ or exp Brain Injuries/ or exp Intracranial Aneurysm/ or exp Intracranial Arterial Diseases/ or exp Cerebrovascular Disorders/ or exp Cerebrovascular Trauma/ or ((operat* or surg* or procedur*) adj5 (brain* or neuro*)).mp. or ((Brain or Cerebro*) adj5 (disord* or trauma* or damage)).mp. or Intracranial.mp.
3. 1 and 2
4. ((randomized controlled trial or controlled clinical trial).pt. or randomized.ab. or placebo.ab. or clinical trials as topic.sh. or randomly.ab. or trial.ti.) not (animals not (humans and animals)).sh.
5. 3 and 4

Appendix 4. Search strategy for EMBASE (OvidSP)

1. exp hypothermia/ or exp induced hypothermia/ or exp profound induced hypothermia/ or exp cryotherapy/ or exp cryosurgery/ or (Cooling or Cryotherapy or Hypotherm*).mp. or (hypotherm* adj5 induced).mp. or (cryo* or cold).ti,ab.
2. exp neurosurgery/ or exp brain injury/ or intracranial aneurysm/ or cerebral artery disease/ or brain hemorrhage/ or exp cerebrovascular accident/ or exp cerebrovascular disease/ or exp cerebrovascular surgery/ or ((operat* or surg* or procedur*) adj5 (brain* or neuro*)).mp. or ((Brain or Cerebro*) adj5 (disord* or trauma* or damage)).mp. or Intracranial.mp.
3. 1 and 2
4. (placebo.sh. or controlled study.ab. or random*.ti,ab. or trial*.ti,ab.) not (animals not (humans and animals)).sh.
5. 3 and 4

Appendix 5. Search strategy for LILACS (BIREME)

"HYPOTHERMIA/" or "CRYOTHERAPY" or "CRYOTHERAPY/" or "cryo$" or "cold$" or "hypotherm$" or "Arrest"

Appendix 6. Search strategy for CINAHL (EBSCOhost)

S1 MH Hypothermia Treatment (Iowa NIC) or MH Hypothermia (NANDA) or MJ Hypothermia or MH Hypothermia, Induced or MH Hypothermia (Saba CCC) or MJ Cryotherapy or MJ Cryosurgery or TX (hypotherm* and induced) or AB cold* or TI cold* or TX cryo*
S2 MJ Neurosurgery or MH Brain Injuries or MH Cerebral Aneurysm or MH Intracranial Hemorrhage or MJ Cerebral Vascular Accident or MH Cerebrovascular Disorders or TX ((operat* or surg* or procedur*) and (brain* or neuro*)) or TX ((Brain or Cerebro*) and (disord* or trauma* or damage)) or TX Intracranial
S3 S1 and S2

Appendix 7. Data Extraction Form

Data Extraction form

Source
Study id
Citation
Eligibility
Confirm reason for eligibility
Methods
Study design
Study duration
Sequence generation
Allocation Concealment
Randomization Method
Blinding

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(Continued)

Participants	
Number	
Age	
Gender	
Co morbidities	
Inclusion Criteria	·
Exclusion Criteria	·
Setting	
Interventions	
No of arms	
Intervention type	·
Timing	
Intervention Integrity	
Method of admin	
Ancillary treatment	
Adjunctive Therapy	·
Outcomes	·
Primary	
Secondary	-
Adverse Events	
Results	
No. of participants allocated to each treatment	·
No. who received each treatment	·
No. who did not receive intended treatment and why	
No. followed up	
No. lost and why	
No. included in the final analysis	

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(Continued)

Analysis
Method
Sample size
Reported effect size
Conclusions
Funding source

Appendix 8. World Federation of Neurological Surgeons score

Grade	Glasgow Coma Score	Motor deficit
1	15	absent
2	13 to 14	absent
3	13 to 14	present
4	7 to 12	present or absent
5	3 to 6	present or absent

Appendix 9. Modified Rankin Score

The scale runs from 0 to 6, running from perfect health without symptoms to death

- 0 - no symptoms.
- 1 - no significant disability. Able to carry out all usual activities, despite some symptoms.
- 2 - slight disability. Able to look after own affairs without assistance, but unable to carry out all previous activities.
- 3 - moderate disability. Requires some help, but able to walk unassisted.
- 4 - moderately severe disability. Unable to attend to own bodily needs without assistance, and unable to walk unassisted.
- 5 - severe disability. Requires constant nursing care and attention, bedridden, incontinent.
- 6 - dead.

WHAT'S NEW

Date	Event	Description
26 January 2015	New search has been performed	This review is an update of a previous Cochrane systematic review of the same title (Milani 2011) which included four studies. The previous authors decided not to update the review so this updated version was conducted by five new authors (IMG, RL,

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Date	Event	Description
		JGB, AGD, MCW). One new ongoing trial (Nuegebauer 2013) was identified but no new studies eligible for inclusion were found.
26 January 2015	New citation required but conclusions have not changed	This updated version includes extensive revisions and updates to the text and layout of the review, a more comprehensive risk of bias assessment of the included studies, a summary of findings table and a quality assessment of the evidence. Our conclusions did not differ from those of the original review in that no evidence was found that perioperative hypothermia significantly reduces mortality or severe disability, or increases complications in patients undergoing neurosurgery.

HISTORY

Protocol first published: Issue 3, 2007
Review first published: Issue 10, 2011

Date	Event	Description
31 May 2012	Amended	Contact details updated.
8 April 2008	Amended	Converted to new review format.

CONTRIBUTIONS OF AUTHORS

This is an updated version of a Cochrane review (Milani 2011) which was originally conceived by Wilson Roberto Oliveira Milani. The original authors (WRO Milani, PL Antibas and GF Prado) also conducted the meta-analysis and as the updated version included no new studies their original analysis was not repeated or modified for this most recent version.

For this updated version, authors' contributions were as follows.

Co-ordination of the revised version - Imelda M Galvin (IMG).

Screening of search results and re-extraction of data from original studies - IMG, Ron Levy (RL) and John Gordon Boyd (JGB).

Risk of bias assessment and study quality assessment for included studies - IMG and Ron Levy (RL).

Writing the report - Imelda M Galvin (IMG).

Statistical expertise - Andrew G Day (AGD) (looked over the methods and analysis and read the final report and confirmed that it was accurate).

Senior content expertise - Michael C Wallace (MCW).

All authors of this updated version contributed to the content of the review and reviewed and agreed on the final version.

DECLARATIONS OF INTEREST

Imelda M Galvin: I have no competing interests related to the present review that need to be declared.

Ron Levy: I have no competing interests related to the present review that need to be declared.

John G Boyd: I have no competing interests related to the present review that need to be declared.

Andrew G Day: I have no competing interests related to the present review that need to be declared.

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Micheal C Wallace: I am on the board of an organization (Canadian Medical Protective Association) promoting safer medical care and providing medical practice protection.

SOURCES OF SUPPORT

Internal sources

- None, Other.

External sources

- None, Other.

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

Differences between the published protocol and Milani 2011.

1. For statistical analysis the RevMan 5.2 program was used.
2. Humberto Saconato was an author of the published protocol.

INDEX TERMS

Medical Subject Headings (MeSH)

Brain [*surgery]; Brain Ischemia [*prevention & control]; Hypothermia, Induced [*methods] [mortality]; Intraoperative Complications [*prevention & control]; Randomized Controlled Trials as Topic

MeSH check words

Humans

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